

AP EAMCET-2019-Engineering-20 Apr-Shift-1
Questions Mathematics

1. $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 2\cos^3 \frac{x}{2}$ on $\mathbb{R} - \{0\}$ is

- (1) One one function
- (2) Bijection
- (3) Algebraic function
- (d) Even function

2. Consider the following lists

List I

A) $f(x) = \frac{|x+2|}{x+2}, x \neq -2$

B) $g(x) = [x], x \in \mathbb{R}$

C) $h(x) = |x - [x]|, x \in \mathbb{R}$

D) $f(x) = \frac{1}{2 - \sin 3x}, x \in \mathbb{R}$

List II

I) $\left[\frac{1}{3}, 1\right]$

(II) $\mathbb{Z}$

(III) $\mathbb{W}$

(IV) $[0, 1)$

(V) $\{-1, 1\}$

	A	B	C	D
(1)	V	III	II	I

	A	B	C	D
(2)	III	II	IV	I

	A	B	C	D
(3)	V	III	IV	I

	A	B	C	D
(4)	I	II	III	IV

3. Assertion (A):

$$(1) + (1 + 2 + 4) + (4 + 6 + 9) + (9 + 12 + 16) + \dots + (81 + 90 + 100) = 1000$$

Reason (R): $\sum_{r=1}^n (r^3 - (r-1)^3) = n^3$ for any natural number n .

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

4. If $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $X = APA^T$ then $A^T X^{50} A =$

(1) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(2) $\begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$

(3) $\begin{bmatrix} 25 & 1 \\ 1 & -25 \end{bmatrix}$

(4) $\begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}$

5. If $[x]$ is the greatest integer less than or equal to x and $|x|$ is the modulus of x , then the system of three equations $2x + 3|y| + 5|z| = 0$, $x + |y| - 2|z| = 4$, $x + |y| + [z] = 1$ has

(1) a unique solution

(2) finitely many solutions

(3) infinitely many solutions

(4) no solutions

6. Investigate the λ and μ for the system $x + 2y + 3z = 6$,
 $x + 3y + 5z = 9$, $2x + 5y + \lambda z = \mu$ and match the following values
 in List I with the items in List II

List I

A) $\lambda = 8, \mu \neq 15$

B) $\lambda \neq 8, \mu \in \mathbb{R}$

C) $\lambda = 8, \mu = 15$

List II

I) Infinitely many solutions

II) No solutions

III) Unique solutions

The correct match is

(1)

A	B	C
I	II	III

(2)

A	B	C
II	III	I

(3)

A	B	C
III	I	II

(4)

A	B	C
III	II	I

7. If $z_n = x + iy$, $x, y \in \mathbb{R}$, $(x, y) \neq (0, -4)$ and $\text{Arg}\left(\frac{2z-3}{z+4i}\right) = \frac{\pi}{4}$

then the locus of z is

(1) $2x^2 + 2y^2 + 5x + 5y - 12 = 0$

$$(2) 2x^2 - 3xy + y^2 + 5x + y - 12 = 0$$

$$(3) 2x^2 + 3xy + y^2 + 5x + y + 12 = 0$$

$$(4) 2x^2 + 2y^2 - 11x + 7y - 12 = 0$$

8. If $z = x + iy$, $x, y \in R$ and the imaginary part of $\frac{\bar{z} - 1}{z - i}$ is 1 then

locus of z is

$$(1) x + y + 1 = 0$$

$$(2) x + y + 1 = 0, (x, y) \neq (0, -1)$$

$$(3) x^2 + y^2 - x + 3y + 2 = 0$$

$$(4) x^2 + y^2 - x + 3y + 2 = 0, (x, y) \neq (0, -1)$$

9. If ω represents a complex cube root of unity, then

$$\left(1 + \frac{1}{\omega}\right)\left(1 + \frac{1}{\omega^2}\right) + \left(2 + \frac{1}{\omega}\right)\left(2 + \frac{1}{\omega^2}\right) + \dots + \left(n + \frac{1}{\omega}\right)\left(n + \frac{1}{\omega^2}\right) =$$

$$(1) \frac{n(n^2 + 1)}{3}$$

$$(2) \frac{n(n^2 + 2)}{3}$$

$$(3) \frac{n(n^2 - 2)}{3}$$

$$(4) \frac{n^2(n-1)}{6}$$

10. If ω represents a complex cube root of unity, then

$$\sum_{r=1}^9 r(r+1-\omega)(r+1-\omega^2) =$$

(1) 5025

(2) 4020

(3) 2016

(4) 3015

11. If α and β are the roots of $x^2 + 7x + 3 = 0$ and $\frac{2\alpha}{3-4\alpha}, \frac{2\beta}{3-4\beta}$

are the roots of $ax^2 + bx + c = 0$ and GCD of a, b, c is 1 then

$$a + b + c =$$

(1) 11

(2) 0

(3) 243

(4) 81

12. If α, β are the roots of $x^2 + bx + c = 0$, γ, δ are the roots of

$$x^2 + b_1x + c_1 = 0 \text{ and } \gamma < \alpha < \delta < \beta \text{ then } (c - c_1)^2 <$$

(1) $(b_1 - b)(bc_1 - b_1c)$

(2) 1

(3) $(b - b_1)^2$

(4) $(c - c_1)(b_1c - b_1c_1)$

13. Let a, b, c be the sides of scalane triangle. If λ is a real number such that root of the equation $x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are real then the interval in which λ lies is

(1) $\left(-\infty, \frac{4}{3}\right)$

(2) $\left(\frac{5}{3}, \infty\right)$

(3) $\left(\frac{1}{3}, \frac{5}{3}\right)$

(4) $\left(\frac{4}{3}, \infty\right)$

14. The polynomial equation of degree 4 having real coefficients with three of its roots as $2 \pm \sqrt{3}$ and $1 + 2i$, is

(1) $x^4 - 6x^3 - 14x^2 + 22x + 5 = 0$

(2) $x^4 - 6x^3 - 19x + 22x - 5 = 0$

(3) $x^4 - 6x^3 + 19x - 22x + 5 = 0$

(4) $x^4 - 6x^3 + 14x^3 - 22x + 5 = 0$

15. All letters of the word ANIMAL are permitted in all possible ways and the permutations thus formed are arranged in dictionary order. If the rank of word ANIMAL is x . Then the permutation with rank x , among the permutation obtained by permuting the letter of the word PERSON and arranging the permutations thus formed in the dictionary order is

(1) ENOPRS

(2) NOSPRE

(3) NOEPRS

(4) ESORNP

16. A student is allowed to choose atmost n books from a collection of $2n + 1$ books. If the total number of ways in which he can select atleast one book is 255, then the value of n is

- (1) 4
- (2) 5
- (3) 6
- (4) 7

17. The sum of all the coefficients in the binomial expansion of $(1 + 2x)^n$ is 6561. Let $R = (1 + 2x)^n = I + F$ where $I \in \mathbb{N}$ and

$$0 < F < 1. \text{ If } x = \frac{1}{\sqrt{2}}, \text{ then } 1 - \frac{F}{1 + (\sqrt{2} - 1)^4} =$$

- (1) $(3\sqrt{2} - 4)$
- (2) $4(3\sqrt{2} + 4)$
- (3) $(\sqrt{2} - 1)^4$
- (4) 1

18. If $\frac{(1 - px)^{-1}}{1 - qx} = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$, then $a_n =$

$$(1) \frac{p^{n+1} - q^{n+1}}{q - p}$$

$$(2) \frac{p^{n+1} - q^{n+1}}{p - q}$$

$$(3) \frac{p^n - q^n}{q - p}$$

$$(4) \frac{p^n - q^n}{p - q}$$

19. If $\frac{3}{(x-1)(x^2+x+1)} = \frac{1}{x-1} - \frac{x+2}{(x^2+x+1)} = f_1(x) - f_2(x)$ and

$$\frac{x+1}{(x-1)^2(x^2+x+1)} = Af_1(x) + \left(B + \frac{D}{x-1}\right)f_2(x) + \frac{C}{(x-1)^2} \quad \text{then}$$

$$A + B + C + D =$$

(1) 1

(2) $-\frac{1}{3}$

(3) 0

(4) $\frac{1}{3}$

20. Let M and m respectively denote the maximum and minimum values of $[f(\theta)]^2$ where

$$f(\theta) = \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} + \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}. \quad \text{Then}$$

$$M - m =$$

(1) $a^2 + b^2$

(2) $(a - b)^2$

(3) $a^2 b^2$

(4) $(a + b)^2$

21. If $\cos A = -\frac{60}{61}$ and $\tan B = -\frac{7}{24}$ and neither A nor B is in the

second quadrant, then the angle $A + \frac{B}{2}$ lies in the quadrant

(1) 1

(2) 2

(3) 3

(4) 4

22. $\cos^2 5^\circ - \cos^2 15^\circ - \sin^2 15^\circ + \sin^2 35^\circ$
 $+ \cos 15^\circ \sin 15^\circ - \cos 5^\circ \sin 35^\circ =$

(1) 0

(2) 1

(3) $\frac{3}{2}$

(4) 2

23. If $\cos \theta \neq 0$ and $\sec \theta - 1 = (\sqrt{2} - 1) \tan \theta$ then $\theta =$

(1) $n\pi + \frac{\pi}{8}, n \in Z$

(2) $2n\pi + \frac{\pi}{4}$ (or) $2n\pi, n \in Z$

(3) $2n\pi + \frac{\pi}{8}, n \in Z$

(4) $2n\pi - \frac{\pi}{4}$ (or) $2n\pi, n \in Z$

24. $\cot \left[\sum_{n=3}^{32} \cot^{-1} \left(1 + \sum_{k=1}^n 2k \right) \right] =$

(1) $\frac{10}{3}$

(2) $\frac{8}{3}$

(3) $\frac{14}{3}$

(4) $\frac{16}{3}$

25. If $\sin x \cos hy = \cos \theta$, $\cos x \sin hy = \sin \theta$ and $4 \tan x = 3$. Then

$\sin h^2 y =$

(1) $\frac{4}{5}$

(2) $\frac{9}{16}$

(3) $\frac{9}{25}$

(4) $\frac{16}{25}$

26. In triangle ABC, if $\frac{b+c}{9} = \frac{c+a}{10} = \frac{a+b}{11} = k$ then

$\frac{\cos A + \cos B}{\cos C} =$

(1) $\frac{9}{10}$

(2) $\frac{10}{11}$

(3) $\frac{11}{12}$

(4) $\frac{12}{13}$

27. In triangle ABC, with usual notation, match the items in list I with the items in list II and choose the correct option

List I

List II

A)

$$r_1 r_2 \sqrt{\left(\frac{4R - r_1 - r_2}{r_1 + r_2} \right)}$$

I)

$$b$$

B)

$$\frac{r_2(r_3 + r_2)}{\sqrt{r_1 r_2 + r_2 r_3 + r_3 r_1}}$$

II)

$$a^2, b^2, c^2 \text{ are in A.P}$$

C)

$$\frac{a}{c} = \frac{\sin(A - B)}{\sin(B - C)}$$

III) Δ

D)

$$bc \cos^2 \frac{A}{2}$$

IV)

$$Rr_1 r_2 r_3$$

V)

$$s(s - a)$$

The correct match is

	A	B	C	D
(1)	IV	III	I	V

	A	B	C	D
(2)	V	IV	III	II

	A	B	C	D
(3)	III	I	II	V

	A	B	C	D
(4)	IV	V	II	I

28. If a, b, c are the sides of $\triangle ABC$ for which $r_1 = 8$, $r_2 = 12$ and $r_3 = 24$ then the ordered triad $(a, b, c) =$

(1) $(12, 20, 16)$

(2) $(12, 16, 20)$

(3) $(16, 12, 20)$

(4) $(20, 16, 12)$

29. If $4\bar{i} + 7\bar{j} + 8\bar{k}$, $2\bar{i} + 3\bar{j} + 4\bar{k}$, $2\bar{i} + 5\bar{j} + 7\bar{k}$ are respectively the position vectors of vertices A, B, C of triangle ABC, then the position vectors of the point where the bisector of angle meets $\overline{BC}$ is

$$(1) \ 2\bar{i} + \frac{13}{3}\bar{j} + 2\bar{k}$$

$$(2) \ 2\bar{i} - \frac{13}{3}\bar{j} + 6\bar{k}$$

$$(3) \ 2\bar{i} + 13\bar{j} + 6\bar{k}$$

$$(4) \ 2\bar{i} + \frac{13}{3}\bar{j} + 6\bar{k}$$

30. The equation of the plane passing through the points $\bar{i} + 2\bar{j} - \bar{k}$ and perpendicular to the line of intersection of the plane $\bar{r} \cdot (3\bar{i} - \bar{j} + \bar{k}) = 1$ and $\bar{r} \cdot (\bar{i} + 4\bar{j} - 2\bar{k}) = 2$ is

$$(1) \ \bar{r} \cdot (-2\bar{i} - 5\bar{j} + \bar{k}) = 0$$

$$(2) \ \bar{r} \cdot (\bar{i} + 7\bar{j} + 4\bar{k}) = 0$$

$$(3) \ \bar{r} \cdot (2\bar{i} - 7\bar{j} - 13\bar{k}) = 1$$

$$(4) \ \bar{r} \cdot (-2\bar{i} + 7\bar{j} + 13\bar{k}) = 0$$

31. If the position vector of the vertices, A, B and C of ΔABC are $\bar{i} + 2\bar{j} - 5\bar{k}$, $-2\bar{i} + 2\bar{j} + \bar{k}$ and $2\bar{i} + \bar{j} - \bar{k}$ respectively then $\angle B =$

$$(1) \cos^{-1}\left(\frac{7}{3\sqrt{10}}\right)$$

$$(2) \cos^{-1}\left(\frac{8}{\sqrt{105}}\right)$$

$$(3) \cos^{-1}\left(\frac{1}{\sqrt{42}}\right)$$

$$(4) \cos^{-1}\left(-\frac{7}{3\sqrt{10}}\right)$$

32. If the position vector of the vertices of ΔABC are $\overrightarrow{OA} = 3\vec{i} + \vec{j} + 2\vec{k}$, $\overrightarrow{OB} = \vec{i} + 2\vec{j} + 3\vec{k}$ and $\overrightarrow{OC} = 2\vec{i} + 3\vec{j} + \vec{k}$ then the length of altitude of triangle ABC drawn from A is

$$(1) \sqrt{\frac{3}{2}}$$

$$(2) \frac{3}{\sqrt{2}}$$

$$(3) \frac{\sqrt{3}}{2}$$

$$(4) \frac{3}{2}$$

33. A new tetrahedron is formed by joining the faces of a given tetrahedron OABC. Then the ratio of volume of new tetrahedron to that of given tetrahedron is

- (1) $\frac{3}{25}$
- (2) $\frac{1}{27}$
- (3) $\frac{5}{62}$
- (4) $\frac{1}{162}$

34. Let $\vec{A} = 2\vec{i} + \vec{j} - 2\vec{k}$ and $\vec{B} = \vec{i} + \vec{j}$ if $\vec{C}$ is such that $\vec{A} \cdot \vec{C} = |\vec{C}|$, $|\vec{C} - \vec{A}| = 2\sqrt{2}$ and the angle between $\vec{A} \times \vec{B}$ and $\vec{C}$ is 30° then the value of $|(\vec{A} \times \vec{C}) \times \vec{C}|$ is

- (1) $\frac{2}{3}$
- (2) $\frac{3}{2}$
- (3) 3
- (4) 2

35. If $a_0, a_1, \dots, a_{11}$ are in an arithmetic progression with common difference d , then their mean deviation from arithmetic mean is

(1) $\frac{30}{11}|d|$

(2) $2|d|$

(3) $3|d|$

(4) $12|d|$

36. The variance of the following continuous frequency distribution is

Class Interval	0-10	10-20	20-30	30-40
Frequency	2	3	4	1

(1) 201

(2) 62

(3) 19

(4) 84

37. If two sections of strength 30 and 45 are formed from 75 students who are admitted in a school, then the probability that two particular students are always together in the same section is

(1) $\frac{66}{185}$

(2) $\frac{19}{37}$

(3) $\frac{29}{185}$

(4) $\frac{18}{37}$

38. A bag contains $2n$ coins out of which $n-1$ are unfair with head on both sides and remaining are fair. One coin is picked from a bag at random and tossed. If the probability that head falls in the

toss is $\frac{41}{56}$ then the number of unfair coins in the bag is

(1) 18

(2) 15

(3) 13

(4) 14

39. Bag A contains 6 Green and 8 Red balls and bag B contains 9 Green and 5 Red balls. A card is drawn at random from a well shuffled pack of 52 playing cards. If it is a spade, two balls are drawn at random from bag A, otherwise two balls are drawn at random from bag B. If the two balls drawn are found to be of the same colour, then the probability that they are drawn from bag A is

(1) $\frac{43}{181}$

(2) $\frac{1}{4}$

(3) $\frac{48}{131}$

(4) $\frac{43}{138}$

40. A random variable X has the probability distribution,

$X = x_i$	1	2	3	4	5	6
$P(X = x_i)$	0.2	0.3	0.12	0.1	0.2	0.08

If $A = \{x_i / x_i \text{ is a prime number}\}$, $B = \{x_i / x_i < 4\}$ are two events then $P(A \cup B) =$

- (1) 0.31
- (2) 0.62
- (3) 0.82
- (4) 0.41

41. In a Poisson distribution with mean $\sum_{x=0}^{\infty} |x - \bar{x}| P(X = x) =$

- (1) e
- (2) $\frac{1}{e}$
- (3) $\frac{2}{e}$
- (4) $\frac{2}{3e}$

42. Two straight rods of length $2a$ and $2b$ move along the coordinate axis in such a way that their extremities are always concyclic. Then the locus of center of such circles is

- (1) $2(x^2 + y^2) = a^2 + b^2$
- (2) $2(x^2 - y^2) = a^2 + b^2$
- (3) $x^2 + y^2 = a^2 + b^2$

(4) $x^2 - y^2 = a^2 - b^2$

43. When the coordinate axes are rotated around the origin in the positive direction through an angle $\frac{\pi}{4}$, if the equation

$25x^2 + 9y^2 = 225$ is transformed to $\alpha x^2 + \beta xy + \gamma y^2 = \delta$ the

$\alpha + \beta + \gamma - (\sqrt{\delta})^2 =$

(1) 3

(2) 9

(3) 4

(4) 16

44. The equation of the line through the point of intersection of the lines $3x - 4y + 1 = 0$ and $5x + y - 1 = 0$ and making equal non-zero intercepts on the coordinate axes is

(1) $2x + 2y = 3$

(2) $23x + 23y = 6$

(3) $23x + 23y = 11$

(4) $2x + 2y = 7$

45. The line through $P(a, 2)$ where $a \neq 0$, making an angle 45° with positive X-axis meets the curve $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at A and D and the coordinate axis at B and C. If PA, PB, PC and PD are in geometric progression then $2a =$

- (1) 13
- (2) 7
- (3) 1
- (4) -13

46. The equation of the perpendicular bisector of the sides AB and AC of a $\triangle ABC$ are $x - y = 5$ and $x + 2y = 0$ respectively. If A is $(1, -2)$ then the equation of straight line BC is

- (1) $14x + 23y - 40 = 0$
- (2) $12x + 17y - 28 = 0$
- (3) $14x - 29y - 30 = 0$
- (4) $7x - 12y + 15 = 0$

47. If each line of a pair of lines original line passing through origin is at a perpendicular distance of 4 units from the point $(3,4)$, then the equation of the pair of lines is

(1) $7x^2 + 24xy = 0$

(2) $7y^2 + 24xy = 0$

(3) $7y^2 - 24xy = 0$

(4) $7x^2 - 24xy = 0$

48. Variable straight lines $y = mx + c$ make intercepts on the curve $y^2 - 4ax = 0$ which subtend a right angle at the origin. Then the point of concurrence of these lines $y = mx + c$ is

(1) $(4a, 0)$

(2) $(2a, 0)$

(3) $(-4a, 0)$

(4) $(-2a, 0)$

49. The abscissae of two points P, Q are the roots of the equation $2x^2 + 4x - 7 = 0$ and their ordinates are the roots of the equation

$3x^2 - 12x - 1 = 0$. Then the centre of the circle with PQ as a diameter is

- (1) $(-1, 2)$
- (2) $(-2, 6)$
- (3) $(1, -2)$
- (4) $(2, -6)$

50. If the angle between a pair of tangents drawn from a point P to the circle $x^2 + y^2 + 4x - 6y + 9\sin^2 \alpha + 13\cos^2 \alpha = 0$ is 2α , then the equation of the locus of P is

- (1) $x^2 + y^2 + 4x - 6y + 4 = 0$
- (2) $x^2 + y^2 + 4x - 6y - 9 = 0$
- (3) $x^2 + y^2 - 4x + 6y - 4 = 0$
- (4) $x^2 + y^2 + 4x - 6y + 9 = 0$

51. The equation of the circle whose radius is 3 and which touches internally the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ at the point $(-1, -1)$ is

- (1) $5x^2 + 5y^2 + 9x - 6y - 7 = 0$

$$(2) 5x^2 + 5y^2 - 8x - 14y - 32 = 0$$

$$(3) 5x^2 + 5y^2 - 6x + 8y - 8 = 0$$

$$(4) 5x^2 + 5y^2 + 6x - 8y - 12 = 0$$

52. Suppose that the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ has its centre on $2x + 3y - 7 = 0$ and cut the circles $x^2 + y^2 - 4x - 6y + 11 = 0$ and $x^2 + y^2 - 10x - 4y + 21 = 0$ orthogonally. Then

$$5g - 10f + 3c =$$

$$(1) 0$$

$$(2) 1$$

$$(3) 3$$

$$(4) 9$$

53. If the radical axis of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ and $2x^2 + 2y^2 + 3x + 8y + 2c = 0$ touches the circle $x^2 + y^2 + 2x + 2y + 1 = 0$ then $(4g - 3)(f - 2) =$

$$(1) 0$$

$$(2) -1$$

$$(3) 1$$

(4) 2

54. The parabola $x^2 = 4ay$ makes an intercept of length $\sqrt{40}$ units on the line $y = 1 + 2x$ then a value of $4a$ is

(1) 2

(2) -2

(3) -1

(4) 2

55. The locus of the point of intersection of perpendicular normals to the parabola $y^2 = ax$ is

(1) $y^2 - 2ax + a^2 = 0$

(2) $y^2 + ax + 2a^2 = 0$

(3) $y^2 - ax + 2a^2 = 0$

(4) $y^2 - ax + 3a^2 = 0$

56. P is a variable point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with foci F_1 and F_2 . If A is the area of the triangle PF_1F_2 , then the maximum value of A is

(1) $\frac{e}{ab}$

(2) $\frac{ae}{b}$

(3) aeb

(4) $\frac{ab}{e}$

57. If the line joining the points $A(\alpha)$ and $B(\beta)$ on the ellipse

$\frac{x^2}{25} + \frac{y^2}{9} = 1$ is a focal chord, then one possible value of

$\cot \frac{\alpha}{2} \cot \frac{\beta}{2}$ is

(1) -3

(2) 3

(3) -9

(4) 9

58. The equation of a tangent to the hyperbola

$16x^2 - 25y^2 - 96x + 100y - 356 = 0$ which makes an angle 45°

with its transverse axis is

(1) $x - y + 2 = 0$

(2) $x - y + 4 = 0$

(3) $x + y + 2 = 0$

(4) $x + y + 4 = 0$

59. If $P(0,7,10)$, $Q(-1,6,6)$ and $R(-4,9,6)$ are three points in the space, then PQR is

(1) Right angled isosceles triangle

(2) Equilateral triangle

(3) Isosceles but not right angled triangle

(4) Scalene triangle

60. $A(2,3,5)$, $B(\alpha,3,3)$ and $C(7,5,\beta)$ are the vertices of a triangle.

If the median through A is equally inclined with the coordinate

axes then $\cos^{-1}\left(\frac{\alpha}{\beta}\right) =$

(1) $\cos^{-1}\left(\frac{-1}{9}\right)$

(2) $\frac{\pi}{2}$

(3) $\frac{\pi}{3}$

(4) $\cos^{-1}\left(\frac{2}{5}\right)$

61. The plane $3x + 4y + 6z + 7 = 0$ is rotated about the line $\vec{r} = (\vec{i} + 2\vec{j} - 3\vec{k}) + t(2\vec{i} - 3\vec{j} + \vec{k})$ until the plane passes through the origin. The equation of the plane in new position is

(1) $x + y + z = 0$

(2) $6x + 3y - 4z = 0$

(3) $4x - 5y - 2z = 0$

(4) $x + 2y + 4z = 0$

62. If $\lim_{x \rightarrow \infty} \left\{ \frac{x^3 + 1}{x^2 + 1} - (\alpha x + \beta) \right\}$ exist and equal to 2 then the ordered pair (α, β) of real numbers is

(1) $(1, -1)$

(2) $(-2, 1)$

(3) $(-1, 1)$

(4) $(1, -2)$

63. For $k > 0$, $\sum_{x=0}^{\infty} \frac{k^x}{x!} \lim_{x \rightarrow \infty} \frac{n!}{(n-x)!} \left(1 - \frac{k}{n}\right)^{n-x} \left(\frac{1}{n}\right)^x =$

- (1) 0
- (2) k
- (3) x
- (4) 1

64. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 5, & \text{if } x \leq 1 \\ a + bx, & \text{if } 1 < x < 3 \\ b + 5x, & \text{if } 3 \leq x < 5 \\ 30, & \text{if } x \geq 5 \end{cases} \text{ then } f \text{ is}$$

- (1) continuous if $a = 5$ and $b = 5$
- (2) continuous if $a = 0$, $b = 5$
- (3) continuous if $a = -5$, $b = 10$
- (4) not continuous for any value of a and b

65. Let $[x]$ denote the greatest integer less than or equal to x . Then the number of points where the function $y = [x] + |1 - x|$, $1 \leq x \leq 3$ is not differentiable, is

- (1) 1
- (2) 2
- (3) 3
- (4) 4

66. If $\sqrt{1-x^6} + \sqrt{1-y^6} = a(x^3 - y^3)$, then $y^2 \frac{d^2 y}{dx^2} =$

- (1) $\sqrt{\frac{1-y^6}{1-x^6}}$
- (2) $x\sqrt{\frac{1-y^6}{1-x^6}}$
- (3) $x^2\sqrt{\frac{1-y^6}{1-x^6}}$
- (4) $\frac{1}{x^2}\sqrt{\frac{1-y^6}{1-x^6}}$

67. If $y = f(x)$ is twice differentiable function such that a point P,

$$\frac{dy}{dx} = 4, \frac{d^2 y}{dx^2} = -3, \text{ then } \left(\frac{d^2 x}{dy^2} \right)_P =$$

- (1) $\frac{64}{3}$

(2) $\frac{16}{3}$

(3) $\frac{3}{16}$

(4) $\frac{3}{64}$

68. The time T of oscillation of a simple pendulum of length L is governed by $T = 2\pi\sqrt{\frac{L}{g}}$, where g is constant. The percentage by which length be changed in order to correct an error of loss equal to 2 minutes of time per day is

(1) $-\frac{5}{18}$

(2) $-\frac{2}{9}$

(3) $\frac{1}{6}$

(4) $\frac{1}{9}$

69. Let A , G , H and S respectively denote the arithmetic mean, geometric mean, harmonic mean and the sum of the numbers $a_1, a_2, a_3, \dots, a_n$. Then the value of x at which the function

$$f(x) = \sum_{k=1}^n (x - a_k)^2 \text{ has minimum is}$$

- (1) S
- (2) H
- (3) G
- (4) A

70. For $m > 1$, $n > 1$, the value of c for which the Rolle's theorem is applicable for the function $f(x) = x^{2m-1}(a-x)^{2n}$ in $(0, a)$ is

- (1) $\frac{2am-1}{m+2n-1}$
- (2) $\frac{a(m-n+1)}{2m+2n}$
- (3) $\frac{a(2m-1)}{2m+2n-1}$
- (4) $\frac{a(2m+1)}{m+n-1}$

71. If the function $f : [-1, 1] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 2^x + 1, & x \in [-1, 0] \\ 1, & x = 0 \\ 2^x - 1, & x \in (0, 1] \end{cases} \quad \text{then in } [-1, 1], f(x) \text{ has}$$

- (1) a maximum
- (2) a minimum
- (3) both maximum and minimum
- (4) neither maximum nor minimum

72. $\int \frac{x-1}{(x+1)\sqrt{x^3+x^2+x}} dx =$

- (1) $2 \tan^{-1} \left(\sqrt{\frac{1+x+x^2}{x}} \right) + c$
- (2) $\tan^{-1} \left(\sqrt{\frac{1+x+x^2}{x}} \right) + c$
- (3) $2 \tan^{-1} \left(\sqrt{\frac{x}{1+x+x^2}} \right) + c$
- (4) $\tan^{-1} \left(\sqrt{\frac{1+x^2}{x}} \right) + c$

73. If $I(x) = \int x^2 (\log x)^2 dx$ and $I(1) = 0$, then $I(x) =$

(1) $\frac{x^3}{18} [8(\log x)^2 - 3\log x] + \frac{7}{18}$

(2) $\frac{x^3}{27} [9(\log x)^2 + 6\log x] - \frac{2}{27}$

(3) $\frac{x^3}{27} [9(\log x)^2 - 6\log x + 2] - \frac{2}{27}$

(4) $\frac{x^3}{27} [9(\log x)^2 - 6\log x - 2] + \frac{2}{27}$

74. $\int \frac{x^5 dx}{(x^2 + x + 1)(x^6 + 1)(x^4 - x^3 + x - 1)} =$

(1) $\log_e \left| \frac{x^6 - 1}{x^6 + 1} \right| + c$

(2) $\frac{1}{12} \log_e \left| \frac{x^6 - 1}{x^6 + 1} \right| + c$

(3) $\frac{1}{12} \log_e \left| \frac{x^4 + 1}{x^4 - 1} \right| + c$

$$(4) \log_e \left| \frac{x^8 + 1}{x^6 - 1} \right| + c$$

$$75. \int \frac{dx}{x + \sqrt{x-1}} =$$

$$(1) \log_e |x + \sqrt{x-1}| - \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2\sqrt{x-1} + 1}{\sqrt{3}} \right) + c$$

$$(2) \frac{1}{\sqrt{3}} \log_e |x + \sqrt{x-1}| - \tan^{-1} \left(\frac{2\sqrt{x-1} + 1}{\sqrt{3}} \right) + c$$

$$(3) \frac{2}{\sqrt{3}} \log_e |x + \sqrt{x-1}| - \tan^{-1} \left(\frac{2\sqrt{x-1} + 1}{\sqrt{3}} \right) + c$$

$$(4) \log_e |x + \sqrt{x-1}| - \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2\sqrt{x-1} + 1}{\sqrt{3}} \right) + c$$

$$76. \int_{\log_e 2}^x \frac{dt}{\sqrt{e^t - 1}} = \frac{\pi}{6} \Rightarrow x =$$

$$(1) 2 \cdot \log_e 2$$

$$(2) 3 \cdot \log_e 2$$

$$(3) 4 \cdot \log_e 2$$

$$(4) 8 \cdot \log_e 2$$

77. $\int_0^1 \frac{\log_e(1+x)}{1+x^2} dx =$

(1) $\frac{\pi}{4} \log_e 2$

(2) $\frac{\pi}{6} \log_e 6$

(3) $\frac{\pi}{2} \log_e 8$

(4) $\frac{\pi}{8} \log_e 2$

78. If the area of the circle $x^2 + y^2 = 2$ is divided into parts by the parabola $y = x^2$, then the area (in sq. units) of the larger part is

(1) $\frac{3\pi}{2} - \frac{1}{3}$

(2) $6\pi - \frac{4}{3}$

(3) $\frac{4\pi}{3} - \frac{2}{3}$

(4) $4\pi - \frac{1}{4}$

79. If c is a parameter, then the differential equation of the family of curves $x^2 = c(y + c)^2$ is

(1) $x\left(\frac{dy}{dx}\right)^3 + y\left(\frac{dy}{dx}\right)^2 - 1 = 0$

(2) $x\left(\frac{dy}{dx}\right)^3 - y\left(\frac{dy}{dx}\right)^2 + 1 = 0$

(3) $x\left(\frac{dy}{dx}\right)^3 + y\left(\frac{dy}{dx}\right)^2 + 1 = 0$

(4) $x\left(\frac{dy}{dx}\right)^3 - y\left(\frac{dy}{dx}\right)^2 - 1 = 0$

80. If $f(x)$, $f'(x)$, $f''(x)$ are positive functions and $f(0) = 1$, $f'(0) = 2$, then the solution of the differential equation

$\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix} = 0$ is

(1) e^{2x}

(2) $2\sin x + 1$

(3) $\sin^2 x + 2x + 1$

(4) e^{4x}

(4) 4 ms^{-1}

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Answer Key

1	4	33	2	65	4	97	4	129	2
2	3	34	2	66	3	98	1	130	4
3	1	35	3	67	4	99	4	131	3
4	4	36	4	68	1	100	3	132	3
5	3	37	2	69	4	101	1	133	1
6	2	38	3	70	3	102	2	134	1
7	1	39	1	71	4	103	3	135	3
8	4	40	3	72	1	104	1	136	2
9	2	41	3	73	3	105	2	137	1
10	4	42	4	74	2	106	4	138	3
11	4	43	2	75	4	107	4	139	4
12	1	44	3	76	1	108	1	140	3
13	1	45	1	77	4	109	3	141	1
14	4	46	1	78	1	110	4	142	2
15	4	47	2	79	4	111	4	143	1

16	1	48	1	80	1	112	2	144	3
17	3	49	1	81	3	113	2	145	2
18	2	50	4	82	3	114	4	146	4
19	3	51	2	83	3	115	3	147	3
20	2	52	4	84	3	116	2	148	3
21	1	53	1	85	1	117	3	149	4
22	1	54	2	86	4	118	2	150	3
23	2	55	4	87	1	119	1	151	4
24	1	56	3	88	3	120	1	152	3
25	4	57	3	89	1	121	2	153	2
26	3	58	1	90	4	122	3	154	3
27	3	59	1	91	2	123	1	155	4
28	2	60	1	92	2	124	2	156	1
29	4	61	1	93	3	125	*	157	4
30	3	62	4	94	1	126	3	158	3
31	2	63	4	95	4	127	4	159	3
32	2	64	4	96	4	128	1	160	2

AP EAMCET-2019-Engineering-20 Apr-Shift-1
Solutions – Mathematics

1. Consider the function,

$$f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 2\cos^3 \frac{x}{2}$$

Then,

$$\begin{aligned} f(-x) &= \frac{-x}{e^{-x} - 1} - \frac{x}{2} + 2\cos^3 \left(-\frac{x}{2} \right) \\ &= \frac{-xe^x}{1 - e^x} - \frac{x}{2} + 2\cos^3 \frac{x}{2} \\ &= \frac{xe^x}{e^x - 1} - \frac{x}{2} + 2\cos^3 \frac{x}{2} \\ &= \frac{x(e^x - 1 + 1)}{e^x - 1} - \frac{x}{2} + 2\cos^3 \frac{x}{2} \end{aligned}$$

Further simplify the above,

$$\begin{aligned} f(-x) &= x + \frac{x}{e^x - 1} - \frac{x}{2} + 2\cos^3 \frac{x}{2} \\ &= \frac{x}{e^x - 1} + \frac{x}{2} + 2\cos^3 \frac{x}{2} \\ &= f(x), \forall x \in R - \{0\} \end{aligned}$$

This implies $f(x)$ is an even function.

2. For A, Consider the function,

$$f(x) = \frac{|x+2|}{x+2}, x \neq -2$$

Then,

$$\begin{aligned} f(x) &= \begin{cases} \frac{x+2}{x+2}, & x > -2 \\ -\frac{x+2}{x+2}, & x < -2 \end{cases} \\ &= \begin{cases} 1, & x > -2 \\ -1, & x < -2 \end{cases} \end{aligned}$$

This implies, the range of $f(x)$ is $\{-1, 1\}$.

For B,

$$g(x) = \lfloor x \rfloor, x \in \mathbb{R}$$

Since $\lfloor x \rfloor \in I$ so $\lfloor x \rfloor \in W$

This implies the range of $g(x)$ is W .

For C,

$$h(x) = |x - \lfloor x \rfloor|, x \in \mathbb{R}$$

$$= \{x\} \in [0, 1)$$

This implies the range is $[0, 1)$.

For D,

$$f(x) = \frac{1}{2 - \sin 3x}, x \in \mathbb{R}$$

Then,

$$-1 \leq \sin 3x \leq 1$$

$$2 - 1 \leq 2 - \sin 3x \leq 2 + 1$$

$$\frac{1}{3} \leq \frac{1}{2 - \sin 2x} \leq \frac{1}{1}$$

This implies the range of $f(x)$ is $\left[\frac{1}{3}, 1\right]$.

3. Consider the expression,

$$(1) + (1 + 2 + 4) + (4 + 6 + 9) + (9 + 12 + 16) + \dots + (81 + 90 + 100)$$

Then,

$$\begin{aligned} & (1) + \left(\begin{matrix} 1^2 \\ + (2 \times 1) + 2^2 \end{matrix} \right) + \left(\begin{matrix} 2^2 \\ + (2 \times 3) + 3^2 \end{matrix} \right) + \left(\begin{matrix} 3^2 + \\ (3 \times 4) + 4^2 \end{matrix} \right) + \dots \\ & + \left(\begin{matrix} 9^2 + \\ (9 \times 10) + 10^2 \end{matrix} \right) \end{aligned}$$

This implies,

$$\begin{aligned}
 \sum_{r=1}^{10} \left[(r-1)^2 + r(r-1) + r^2 \right] &= \sum_{r=1}^{10} \left[r - (r-1)(r-1)^2 + r(r-1) + r^2 \right] \\
 &= \sum_{r=1}^{10} r^3 - (r-1)^3 \\
 &= (1^3 - 0^3) + (2^3 - 1^3) + (3^3 - 2^3) + \dots + \\
 &\quad (10^3 - 9^3) \\
 &= 10^3 - 0^3
 \end{aligned}$$

Further simplify the above,

$$\sum_{r=1}^{10} \left[(r-1)^2 + r(r-1) + r^2 \right] = 1000$$

This implies, both (A) and (R) are true and (R) is the correct explanation of (A).

4. Consider the matrix,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Since above matrix satisfies $AA^T = I$ so, it is an orthogonal matrix then,

$$\begin{aligned}
 A^T X^{50} A &= A^T X^{49} (A P A^T) A \\
 &= A^T X^{49} A P (A^T A) \\
 &= A^T X^{49} A P \\
 &= A^T X^{48} A P^2 \dots
 \end{aligned}$$

Further simplify the above,

$$\begin{aligned}
 A^T X^{50} A &= A^T A P^{50} \\
 &= I P^{50} \\
 &= P^{50}
 \end{aligned}$$

This implies,

$$\begin{aligned}
 P &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \\
 P^2 &= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \\
 P^3 &= \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \\
 &\vdots \\
 P^{50} &= \begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}
 \end{aligned}$$

Therefore, the correct option is (4).

5. Consider the system of equations,

$$2x + 3|y| + 5[z] = 0$$

$$x + |y| - 2[z] = 4$$

$$x + |y| + [z] = 1$$

Then by Cramer's rule,

$$x = \frac{\Delta_1}{\Delta}$$

$$|y| = \frac{\Delta_2}{\Delta}$$

$$[z] = \frac{\Delta_3}{\Delta}$$

Where,

$$\Delta = \begin{vmatrix} 2 & 3 & 5 \\ 1 & 1 & -2 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= 2(1+2) - 3(1+2) + 5(1-1)$$

$$= -3$$

and,

$$\Delta_1 = \begin{vmatrix} 0 & 3 & 5 \\ 4 & 1 & -2 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= 0(1+2) - 3(4+2) + 5(4-1)$$

$$= -3$$

and,

$$\begin{aligned}\Delta_2 &= \begin{vmatrix} 2 & 0 & 5 \\ 1 & 4 & -2 \\ 1 & 1 & 1 \end{vmatrix} \\ &= 2(4+2) - 0(1+2) + 5(1-4) \\ &= -3\end{aligned}$$

and,

$$\begin{aligned}\Delta_3 &= \begin{vmatrix} 2 & 3 & 0 \\ 1 & 1 & 4 \\ 1 & 1 & 1 \end{vmatrix} \\ &= 2(1-4) - 3(1-4) + 0(1-1) \\ &= 3\end{aligned}$$

This implies,

$$\begin{aligned}x &= \frac{-3}{-3} \\ &= 1\end{aligned}$$

And,

$$\begin{aligned}|y| &= \frac{-3}{-3} \\ &= 1\end{aligned}$$

And,

$$\begin{aligned}[z] &= \frac{-3}{3} \\ &= -1\end{aligned}$$

This implies,

$$z \in [-1, 0)$$

So, the system has infinitely many solutions.

6. Consider the system of equations

$$x + 2y + 3z = 6$$

$$x + 3y + 5z = 9$$

$$2x + 5y + \lambda z = \mu$$

Now from the Cramer's rule,

$$\begin{aligned}\Delta &= \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & \lambda \end{vmatrix} \\ &= 1(3\lambda - 25) - 2(\lambda - 10) + 3(5 - 6) \\ &= \lambda - 8\end{aligned}$$

And,

$$\begin{aligned}\Delta_1 &= \begin{vmatrix} 6 & 2 & 3 \\ 9 & 3 & 5 \\ \mu & 5 & \lambda \end{vmatrix} \\ &= 6(3\lambda - 25) - 2(9\lambda - 5\mu) + 3(45 - 3\mu) \\ &= \mu - 15\end{aligned}$$

And,

$$\begin{aligned}\Delta_2 &= \begin{vmatrix} 1 & 6 & 3 \\ 1 & 9 & 5 \\ 2 & \mu & \lambda \end{vmatrix} \\ &= 1(9\lambda - 5\mu) - 6(\lambda - 10) + 3(\mu - 18) \\ &= 3\lambda - 2\mu + 6\end{aligned}$$

And,

$$\begin{aligned}\Delta_3 &= \begin{vmatrix} 1 & 2 & 6 \\ 1 & 3 & 9 \\ 2 & 5 & \mu \end{vmatrix} \\ &= 1(3\mu - 45) - 2(\mu - 18) + 6(5 - 6) \\ &= \mu - 15\end{aligned}$$

Now, if $\lambda = 8$ and $\mu \neq 15$, then system of linear equation has no solution.

If $\lambda \neq 8$ and $\mu \in \mathbb{R}$, then system of linear equation has a unique solution.

If $\lambda = 8$ and $\mu = 15$, then system of linear equation has infinite number of solutions. So, $\Delta_2 = 3\lambda - 2\mu + 6$ is also zero therefore, the correct option is (2).

7. Consider the expression,

$$z = x + iy, \quad x, y \in R, \quad (x, y) \neq (0, -4)$$

This implies,

$$\begin{aligned} \frac{2z-3}{z+4i} &= \frac{(2x-3)+2iy}{x+i(y+4)} \times \frac{x-i(y+4)}{x-i(y+4)} \\ &= \frac{(2x^2-3x+2y^2+8y)+i(2xy-2xy+3y-8x+12)}{x^2+(y+4)^2} \\ &= \frac{(2x^2-3x+2y^2+8y)+i(12+3y-8x)}{x^2+(y+4)^2} \end{aligned}$$

Then,

$$\begin{aligned} \text{Arg}\left(\frac{2z-3}{z+4i}\right) &= \text{Tan}^{-1}\left(\frac{12+3y-8x}{2x^2-3x+2y^2+8y}\right) \\ &= \frac{\pi}{4} \end{aligned}$$

This implies,

$$\frac{12 + 3y - 8x}{2x^2 - 3x + 2y^2 + 8y} = 1$$

$$2x^2 + 2y^2 + 5x + 5y - 12 = 0$$

8. The given expression is,

$$z = x + iy$$

Then,

$$\begin{aligned} \frac{\bar{z} - 1}{z - i} &= \frac{x - iy - 1}{x - iy - 1} \times \frac{x + i(y + 1)}{x + i(y + 1)} \\ &= \frac{[x(x - 1) + y(y + 1)] + i[(y + 1)(x - 1) - xy]}{x^2 + (y + 1)^2} \\ \operatorname{Im}\left(\frac{\bar{z} - 1}{z - i}\right) &= \frac{xy - y + x - 1 - xy}{x^2 + (y + 1)^2} \\ &= 1 \end{aligned}$$

This implies,

$$x^2 + y^2 - x + 3y + 2 = 0, (x, y) \neq (0, -1).$$

9. Consider the expression,

$$1 + \omega + \omega^2 = 0$$

And,

$$\omega^3 = 1$$

Then,

$$\begin{aligned}\left(r + \frac{1}{\omega}\right)\left(r + \frac{1}{\omega^2}\right) &= r^2 + r\left(\frac{1}{\omega} + \frac{1}{\omega^2}\right) + \frac{1}{\omega^3} \\ &= r^2 + \left(\frac{\omega^2 + \omega}{\omega^3}\right)r + \frac{1}{\omega^3} \\ &= r^2 - r + 1\end{aligned}$$

Then,

$$\begin{aligned}\sum_{r=1}^n \left(r + \frac{1}{\omega}\right)\left(r + \frac{1}{\omega^2}\right) &= \sum_{r=1}^n (r^2 - r + 1) \\ &= \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + n \\ &= \frac{n(n+1)}{6} [2n+1-3] + n \\ &= \frac{n(n+1)}{6} (2n-2) + n\end{aligned}$$

Further simplify the above,

$$\begin{aligned}\sum_{r=1}^n \left(r + \frac{1}{\omega}\right)\left(r + \frac{1}{\omega^2}\right) &= \frac{n}{3} (n^2 - 1) + n \\ &= \frac{n}{3} [n^2 - 1 + 3] \\ &= \frac{n(n^2 + 2)}{3}\end{aligned}$$

10. Since ω is the complex cube root of unity then,

$$\begin{aligned} r(r+1-\omega)(r+1-\omega^2) &= r\left[(r+1)^2 - (\omega + \omega^2)(r+1) + \omega^3\right] \\ &= r\left[(r+1)^2 + (r+1) + 1\right] \\ &= r(r^2 + 3r + 3) \\ &= r^3 + 3r^2 + 3r \end{aligned}$$

This implies,

$$\begin{aligned} \sum_{r=1}^9 r(r+1-\omega)(r+1-\omega^2) &= \sum_{r=1}^9 (r^3 + 3r^2 + 3r) \\ &= \left(\frac{9(10)}{2}\right)^2 + 3\left(\frac{9(10)(19)}{6}\right) + 3\left(\frac{9 \times 10}{2}\right) \\ &= 45^2 + (45)(19) + (3)(45) \\ &= 3015 \end{aligned}$$

11. Let consider,

$$y = \frac{2\alpha}{3-4\alpha}$$

Then,

$$2\alpha = 3y - 4\alpha y$$

$$\alpha = \frac{3y}{2+4y}$$

Here, α is the root of equation $x^2 + 7x + 3 = 0$

So,

$$\left(\frac{3y}{2+4y}\right)^2 + 7\left(\frac{3y}{2+4y}\right) + 3 = 0$$

$$9y^2 + 84y^2 + 42y + 48y^2 + 48y + 12 = 0$$

$$141y^2 + 90y + 12 = 0$$

$$47y^2 + 30y + 4 = 0$$

So,

$$ax^2 + bx + c = 0$$

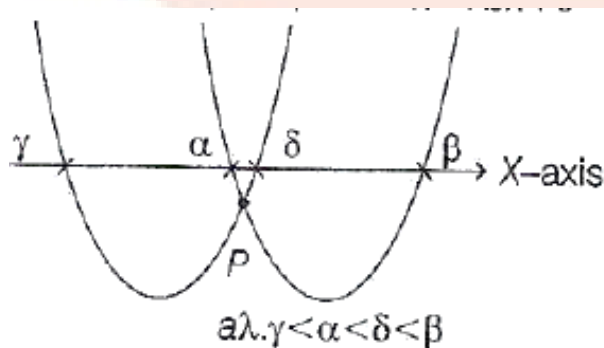
Comparison gives,

$$a = 47, b = 30 \text{ and } c = 4$$

The GCD of 47, 30, 4 is 1. So,

$$\begin{aligned} a + b + c &= 47 + 30 + 4 \\ &= 81 \end{aligned}$$

12. Consider the diagram,



For x coordinate of P on subtraction of given quadratic equations,

$$x^2 + b_1x + c_1 = 0$$

$$x^2 + bx + c = 0$$

Then,

$$(b_1 - b)x + (c_1 - c) = 0$$

$$x = \frac{c - c_1}{b_1 - b}$$

This implies,

$$f(x) = x^2 + bx + c$$

$$f\left(\frac{c - c_1}{b_1 - b}\right) < 0$$

$$\left(\frac{c - c_1}{b_1 - b}\right)^2 + b\left(\frac{c - c_1}{b_1 - b}\right) + c < 0$$

$$(c - c_1)^2 < b(c - c_1)(b_1 - b) - c(b_1 - b)^2$$

Then,

$$(c - c_1)^2 < (b_1 - b)[bc_1 - bc - cb_1 + cb]$$

$$(c - c_1)^2 < (b_1 - b)(bc_1 - cb_1)$$

13. The roots of the given equation are real so,

$$D \geq 0$$

This implies,

$$4(a+b+c)^2 - 4 \times 3\lambda(ab+bc+ca) \geq 0$$

$$(a+b+c)^2 - 3\lambda(ab+bc+ca) \geq 0$$

$$\lambda \leq \frac{(a+b+c)^2}{3(ab+bc+ca)}$$

For scalene triangle,

$$|b-c| < a, |c-a| < b, |a-b| < c$$

So,

$$(b-c)^2 + (c-a)^2 + (a-b)^2 < a^2 + b^2 + c^2$$

$$a^2 + b^2 + c^2 < 2(ab+bc+ca)$$

$$a^2 + b^2 + c^2 + 2(ab+bc+ca) < a^2 + b^2 + c^2 + 4(ab+bc+ca)$$

$$\frac{(a+b+c)^2}{3(ab+bc+ca)} < \frac{4}{3}$$

This implies,

$$\lambda < \frac{4}{3}$$

14. The quadratic equation that has root $2 \pm \sqrt{3}$ is,

$$x^2 + 4x - 1$$

And the equation with root $1 + 2i$ is,

$$x^2 - 2x + 5 = 0$$

So, the required polynomial equation is,

$$(x^2 - 4x + 1)(x^2 - 2x + 5) = 0$$

$$x^4 - 6x^3 + 14x^2 - 22x + 5 = 0$$

15. When the word ANIMAL is arranged in alphabetical order, we get AAILMN, then rank of word ANIMAL is,

$$AA.....4!$$

$$AI.....4!$$

$$AL.....4!$$

$$AM.....4!$$

$$ANA.....3!$$

$$ANIA.....2!$$

$$ANIL.....2!$$

$$ANIMAL.....1$$

Then rank is $x = 107$.

Similarly for word PERSON the alphabetical arrangement is ESORNP then rank of PERSON is,

$EN.....4!$

$EO.....4!$

$EP.....4!$

$ER.....4!$

$ESN.....3!$

$ESON.....2!$

$ESOP.....2!$

$ESORNP.....1$

Then rank is $x = 107$.

16. Consider the expression,

$${}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_n = x$$

Then,

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + \dots + {}^{2n+1}C_n + \dots + {}^{2n+1}C_{2n+1} = 2^{2n+1}$$

$$2x + {}^{2n+1}C_0 + {}^{2n+1}C_{2n+1} = 2^{2n+1}$$

$$2x + 1 + 1 = 2^{2n+1}$$

$$x = 2^{2n} - 1$$

Further simplify the above,

$$2^{2n} - 1 = 255$$

$$2^{2n} = 256$$

$$2^{2n} = 2^8$$

$$n = 4$$

17. It is given that, expansion of $(1+2x)^n$ for $x=1$ is

$$(1+2)^n = 6561$$

$$3^n = 6561$$

$$n = 8$$

$$\text{For } x = \frac{1}{\sqrt{2}}$$

$$R = (1+2x)^n \\ = I + F$$

$$(\sqrt{2}-1)^8 = F'$$

$$(\sqrt{2}+1)^8 + (\sqrt{2}-1)^8 = I + (F + F')$$

This implies,

$$2 \left[(\sqrt{2})^8 \right] + {}^8C_2 (\sqrt{2})^6 + {}^8C_4 (\sqrt{2})^4 + {}^8C_6 (\sqrt{2})^2 + {}^8C_8 \\ = I + (F + F')$$

So,

$$F + F' = 1$$

$$F = 1 - F'$$

$$= 1 - (\sqrt{2}-1)^8$$

So,

$$\begin{aligned}
 1 - \frac{F}{1 + (\sqrt{2} - 1)^4} &= 1 - \frac{1 - (\sqrt{2} - 1)^8}{1 + (\sqrt{2} - 1)^4} \\
 &= 1 - \frac{\left[1 + (\sqrt{2} - 1)^4\right] \left[1 - (\sqrt{2} - 1)^4\right]}{1 + (\sqrt{2} - 1)^4} \\
 &= (\sqrt{2} - 1)^4
 \end{aligned}$$

18. Consider the expression,

$$\frac{(1 - px)^{-1}}{1 - qx} = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$$

$$(1 - px)^{-1} = 1 + px + p^2x^2 + p^3x^3 + \dots + p^nx^n + \dots$$

And,

$$(1 - qx)^{-1} = 1 + qx + q^2x^2 + q^3x^3 + \dots + q^nx^n + \dots$$

Then coefficient of x^n in the expansion of

$$(1 - px)^{-1} (1 - qx)^{-1} = p^n + p^{n-1}q + p^{n-2}q^2 + p^{n-3}q^3 + \dots + q^n$$

Then,

$$\begin{aligned}
 a_n &= \frac{p^n \left(1 - \left(\frac{q}{p} \right)^{n+1} \right)}{1 - \frac{q}{p}} \\
 &= \frac{p^n (p^{n+1} - q^{n+1}) p}{(p - q) p^{n+1}} \\
 &= \frac{p^{n+1} - q^{n+1}}{p - q}
 \end{aligned}$$

19. Consider the expression,

$$\begin{aligned}
 \frac{3}{(x-1)(x^2+x+1)} &= \frac{1}{x-1} - \frac{x+2}{(x^2+x+1)} \\
 &= f_1(x) - f_2(x)
 \end{aligned}$$

And

$$\frac{x+1}{(x-1)^2(x^2+x+1)} = Af_1(x) + \left(B + \frac{D}{x-1} \right) f_2(x) + \frac{C}{(x-1)^2}$$

From above,

$$f_1(x) = \frac{1}{x-1}$$

$$f_2(x) = \frac{x+2}{x^2+x+1}$$

So,

$$\frac{x+1}{(x-1)^2(x^2+x+1)} = \frac{A}{x-1} + \frac{(Bx-B+D)(x+2)}{(x^2+x+1)(x-1)} + \frac{C}{(x-1)^2}$$

$$x+1 = A(x-1)(x^2+x+1) + B(x-1)$$

$$(x+2)(x-1) + D(x+2)(x-1) + C(x^2+x+1)$$

Comparing the coefficients,

$$A + B = 0$$

And,

$$A - A + 2B - 2B + D + C = 0$$

$$D + C = 0$$

This implies $A + B + C + D = 0$.

20. Consider the expression.

$$f(\theta) = \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} + \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$[f(\theta)]^2 = \left(\begin{array}{l} a^2 \cos^2 \theta + b^2 \sin^2 \theta + a^2 \sin^2 \theta + b^2 \cos^2 \theta \\ + 2\sqrt{(a^2 \cos^2 \theta + b^2 \sin^2 \theta)(a^2 \sin^2 \theta + b^2 \cos^2 \theta)} \end{array} \right)$$

The above is maximum when $\sin^2 \theta = \cos^2 \theta = \frac{1}{2}$ and minimum

when these are zero.

So,

$$\begin{aligned} M &= (a^2 + b^2) + 2\sqrt{\left(\frac{a^2}{2} + \frac{b^2}{2}\right)^2} \\ &= 2(a^2 + b^2) \end{aligned}$$

And,

$$\begin{aligned} m &= (a^2 + b^2) + 2\sqrt{a^2 b^2} \\ &= a^2 + b^2 + 2ab \end{aligned}$$

This implies,

$$\begin{aligned} M - m &= 2(a^2 + b^2) - [a^2 + b^2 + 2ab] \\ &= a^2 + b^2 - 2ab \\ &= (a - b)^2 \end{aligned}$$

21. Consider the expression,

$$\tan B = -\frac{7}{24}$$

Then,

$$\frac{2 \tan \frac{B}{2}}{1 - \tan^2 \frac{B}{2}} = -\frac{7}{24}$$

$$7 \tan^2 \frac{B}{2} - 48 \tan \frac{B}{2} - 7 = 0$$

$$\left(7 \tan \frac{B}{2} + 1\right) \left(\tan \frac{B}{2} - 7\right) = 0$$

$$\tan \frac{B}{2} = -\frac{1}{7}$$

Then,

$$\tan \left(A + \frac{B}{2}\right) = \frac{\tan A + \tan \frac{B}{2}}{1 - \tan A \tan \frac{B}{2}}$$

$$\begin{aligned} &= \frac{\frac{11}{60} - \frac{1}{7}}{1 + \frac{11}{420}} \\ &= \frac{17}{431} \end{aligned}$$

This implies,

$$A \in \text{III}^{\text{rd}} \text{ quadrant and } \frac{B}{2} \in \left(\frac{3\pi}{4}, \pi\right)$$

Now,

$\tan\left(A + \frac{B}{2}\right)$ is positive so,

$$\left(A + \frac{B}{2}\right) \in \text{Ist quadrant}$$

22. Consider the expression,

$$\begin{aligned} & \cos^2 5^\circ - \cos^2 15^\circ - \sin^2 15^\circ + \sin^2 35^\circ + \cos^2 15^\circ \sin 15^\circ - \cos 5^\circ \sin 35^\circ \\ &= \cos 5^\circ (\cos 5^\circ - \sin 35^\circ) - \cos 15^\circ (\cos 15^\circ - \sin 15^\circ) + \sin^2 35^\circ - \sin^2 15^\circ \\ &= \cos 5^\circ (\cos 5^\circ - \cos 55^\circ) - \cos 15^\circ (\cos 15^\circ - \cos 75^\circ) + \sin^2 50^\circ \sin^2 20^\circ \\ &= \cos 5^\circ (2 \sin 30^\circ \sin 25^\circ) - \cos 15^\circ (2 \sin 45^\circ \sin 30^\circ) + \sin 50^\circ \sin 20^\circ \\ &= \left(-\frac{\cos 15^\circ}{\sqrt{2}} + \cos 5^\circ \sin 25^\circ \right) \\ &\quad + \sin 50^\circ \sin 20^\circ \\ &= \left(-\frac{\cos 15^\circ}{\sqrt{2}} + \frac{1}{2} (2 \cos 5^\circ \cos 65^\circ) + \frac{1}{2} (2 \sin 50^\circ \sin 20^\circ) \right) \\ &= \left(-\frac{\cos 15^\circ}{\sqrt{2}} + \frac{1}{2} (\cos 70^\circ + \cos 60^\circ + \cos 30^\circ - \cos 70^\circ) \right) \\ &= -\frac{\cos 15^\circ}{\sqrt{2}} + \frac{1}{2} [2 \cos 45^\circ \cos 15^\circ] \\ &= -\frac{\cos 15^\circ}{\sqrt{2}} + \frac{\cos 15^\circ}{\sqrt{2}} \\ &= 0 \end{aligned}$$

23. If $\cos \theta \neq 0$

And,

$$\sec \theta - 1 = (\sqrt{2} - 1) \tan \theta$$

$$\frac{1 - \cos \theta}{\cos \theta} = (\sqrt{2} - 1) \frac{\sin \theta}{\cos \theta}$$

$$2 \sin^2 \frac{\theta}{2} = (\sqrt{2} - 1) 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$\sin \frac{\theta}{2} = 0 \text{ or } \tan \frac{\theta}{2} = \sqrt{2} - 1$$

This implies,

$$\frac{\theta}{2} = n\pi, n \in \mathbb{Z}$$

$$\frac{\theta}{2} = n\pi + \frac{\pi}{8}, n \in \mathbb{Z}$$

$$\theta = 2n\pi + \frac{\pi}{4}$$

$$2n\pi, n \in \mathbb{Z}$$

24. Consider the expression,

$$\sum_{k=1}^n 2k = n(n+1)$$

Then,

$$\cot\left(\sum_{n=3}^{32} \cot^{-1}(1+n(n+1))\right) = \sum_{n=3}^{32} \tan^{-1}\left(\frac{(n+1)-n}{1+(n+1)n}\right)$$

This implies,

$$\begin{aligned} \sum_{n=3}^{32} \tan^{-1}\left(\frac{(n+1)-n}{1+(n+1)n}\right) &= \sum_{n=3}^{32} \tan^{-1}(n+1) - \tan^{-1} n \\ &= (\tan^{-1} 4 - \tan^{-1} 3 + (\tan^{-1} 5 - \tan^{-1} 4)) + \dots \\ &\quad + (\tan^{-1} 33 - \tan^{-1} 32) \\ &= \tan^{-1} 33 - \tan^{-1} 3 \\ &= \tan^{-1} \left[\frac{33-3}{1+99} \right] \end{aligned}$$

Further simplify the above,

$$\begin{aligned} \sum_{n=3}^{32} \tan^{-1}\left(\frac{(n+1)-n}{1+(n+1)n}\right) &= \tan^{-1}\left(\frac{3}{10}\right) \\ \cot\left(\sum_{n=3}^{32} \tan^{-1}\left(\frac{(n+1)-n}{1+(n+1)n}\right)\right) &= \cot\left(\tan^{-1}\left(\frac{3}{10}\right)\right) \\ \cot\left[\cot^{-1}\left(\frac{10}{3}\right)\right] &= \frac{10}{3} \end{aligned}$$

25. Consider the expression,

$$\tan x = \frac{3}{4}$$

Then,

$$\sin^2 x = \frac{9}{25}$$

And

$$\cos^2 x = \frac{16}{25}$$

Then,

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$(\sin \cosh y)^2 + (\cos x \sinh y)^2 = 1$$

$$9 + 9 \sinh^2 y + 16 \sinh^2 y = 25$$

$$\sinh^2 y = \frac{16}{25}$$

26. Let,

$$\frac{b+c}{9} = \frac{c+a}{10} = \frac{a+b}{11} = k$$

This implies,

$$a = 6k, b = 5k, c = 4k$$

Then,

$$\begin{aligned}
 \frac{\cos A + \cos B}{\cos C} &= \frac{\frac{b^2 + c^2 - a^2}{2bc} + \frac{a^2 + c^2 - b^2}{2ac}}{\frac{a^2 + b^2 - c^2}{2ab}} \\
 &= \frac{\frac{25 + 16 - 36}{40} + \frac{36 + 16 - 25}{48}}{\frac{36 + 25 - 16}{60}} \\
 &= \frac{\frac{5}{40} + \frac{27}{48}}{\frac{45}{60}} \\
 &= \frac{11}{12}
 \end{aligned}$$

27. For A,

Consider $r_1 r_2 \sqrt{\frac{4R - r_1 - r_2}{r_1 + r_2}}$

$$= \left[\frac{\Delta^2}{(s-a)(s-b)} \times \left(\sqrt{\frac{4R - 4R \cos \frac{C}{2} \left(\sin \frac{A}{2} \cos \frac{B}{2} + \sin \frac{B}{2} \cos \frac{A}{2} \right)}{4R \cos \frac{C}{2} \left(\sin \frac{A}{2} \cos \frac{B}{2} + \sin \frac{B}{2} \cos \frac{A}{2} \right)}} \right) \right]$$

$$= \frac{\Delta^2}{(s-a)(s-b)} \sqrt{\frac{4R \left(1 - \cos^2 \frac{C}{2} \right)}{4R \cos^2 \frac{C}{2}}}$$

$$= \frac{\Delta^2}{(s-a)(s-b)} \tan \frac{C}{2}$$

$$= \frac{\Delta^2}{(s-a)(s-b)} \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

Then,

$$\begin{aligned} r_1 r_2 \sqrt{\frac{4R - r_1 - r_2}{r_1 + r_2}} &= \frac{\Delta^2}{\Delta} \\ &= \Delta \end{aligned}$$

For B,

$$\begin{aligned}
 \frac{r_2(r_3 + r_2)}{\sqrt{r_1 r_2 + r_2 r_3 + r_3 r_1}} &= \frac{\frac{\Delta}{s-b} \left(\frac{\Delta}{s-c} + \frac{\Delta}{s-a} \right)}{\sqrt{\frac{(s-a) + (s-b) + (s-c)}{(s-a)(s-b)(s-c)}}} \\
 &= \frac{\frac{\Delta}{s-b} \left(\frac{2S - a - c}{(s-a)(s-c)} \right)}{\sqrt{\frac{3S - a - b - c}{(s-a)(s-b)(s-c)}}} \\
 &= \frac{\Delta(b)}{\sqrt{s(s-a)(s-b)(s-c)}} \\
 &= b
 \end{aligned}$$

For C,

$$\begin{aligned}
 \frac{a}{c} &= \frac{\sin(A-B)}{\sin(B-C)} \\
 \frac{\sin A}{\sin C} &= \frac{\sin(A-B)}{\sin(B-C)}
 \end{aligned}$$

$$\sin A \sin(B-C) = \sin(A-B) \sin C$$

$$2 \sin A \cos B \sin C = \sin A \sin B \cos C + \sin B \cos A \sin C$$

This implies,

$$2 \frac{a}{2R} \left(\frac{a^2 + c^2 - b^2}{2ac} \right) \left(\frac{c}{2R} \right) = \left(\frac{a}{2R} \left(\frac{b}{2R} \right) \left(\frac{a^2 + b^2 - c^2}{2abc} \right) \right) +$$

$$\left(\frac{b}{2R} \left(\frac{b^2 + c^2 - a^2}{2bc} \right) \left(\frac{c}{2R} \right) \right)$$

$$2(a^2 + c^2 - b^2) = a^2 + b^2 - c^2 + b^2 + c^2 - a^2$$

$$2a^2 + 2c^2 - 2b^2 = 2b^2$$

$$2b^2 = a^2 + c^2$$

Therefore the correct option is (3).

28. It is given that,

$$r_1 = 8 = \frac{\Delta}{s - a}$$

$$r_2 = 12 = \frac{\Delta}{s - b}$$

$$r_3 = 24 = \frac{\Delta}{s - c}$$

From first two equations,

$$\frac{s - b}{s - a} = \frac{2}{3}$$

$$3s - 3b = 2s - 2a$$

$$5a + c = 5b \quad (I)$$

From r_2 and r_3 ,

$$\frac{s-c}{s-b} = \frac{1}{2}$$

$$2s - 2c = s - b$$

$$a + 3b = 3c \quad (II)$$

From r_1 and r_3 ,

$$\frac{s-c}{s-a} = \frac{1}{3}$$

$$3s - 3c = s - a$$

$$2a + b = 2c \quad (III)$$

From (I), (II), and (III),

$$(a, b, c) = (12, 16, 20)$$

29. The bisector of angle A divides the BC in ratio $c : b$, where

$$c = AB$$

$$= \sqrt{4 + 16 + 16}$$

$$= 6$$

And,

$$b = AC$$

$$= \sqrt{4 + 4 + 1}$$

$$= 3$$

The position vector of the point where bisector of A meet BC is,

$$\frac{6(2\bar{i} + 5\bar{j} + 7\bar{k}) + 3(2\bar{i} + 3\bar{j} + 4\bar{k})}{6 + 3} = \frac{18\bar{i} + 39\bar{j} + 54\bar{k}}{9}$$

$$= 2\bar{i} + \frac{13}{3}\bar{j} + 6\bar{k}$$

30. The equation of the plane passes through the point $(\bar{i} + 2\bar{j} - \bar{k}) = (1, 2, -1)$ having direction ratios normal to a, b, c is,

$$a(x - 1) + b(y - 2) + c(z + 1) = 0 \quad (I)$$

The plane in equation (I) is perpendicular to line of intersection

of planes $\bar{r} \cdot (3\bar{i} - \bar{j} + \bar{k}) = 1$ and $\bar{r} \cdot (\bar{i} + 4\bar{j} - 2\bar{k}) = 2$ so,

$$3a - b + c = 0$$

$$a + 4b - 2c = 0$$

This implies,

$$\frac{a}{2 - 4} = \frac{-b}{-6 - 1} = \frac{c}{12 + 1}$$

$$\frac{a}{2} = \frac{b}{-7} = \frac{c}{-13}$$

So, equation of required plane is,

$$2(x-1) - 7(y-2) - 13(z+1) = 0$$

$$2x - 7y - 13z - 2 + 14 - 13 = 0$$

$$2x - 7y - 13z = 1$$

$$\vec{r} \cdot (2\vec{i} - 7\vec{j} - 13\vec{k}) = 1$$

31. Consider,

$$\overrightarrow{BA} = 3\vec{i} - 6\vec{k}$$

And,

$$\overrightarrow{BC} = 4\vec{i} - \vec{j} - 12\vec{k}$$

Then,

$$\begin{aligned}\cos(\angle B) &= \frac{|\overrightarrow{BA} \cdot \overrightarrow{BC}|}{|\overrightarrow{BA}| |\overrightarrow{BC}|} \\ &= \frac{12 + 12}{\sqrt{9 + 36} \sqrt{16 + 1 + 4}} \\ &= \frac{8}{\sqrt{105}}\end{aligned}$$

$$\angle B = \cos^{-1}\left(\frac{8}{\sqrt{105}}\right)$$

32. The length of altitude drawn from A is given by,

$$h = \frac{\frac{1}{2}|\overline{AB} \times \overline{AC}|}{\frac{1}{2}|\overline{BC}|}$$

$$\overline{AB} = -2\bar{i} + \bar{j} + \bar{k}$$

$$\overline{AC} = -\bar{i} + 2\bar{j} - \bar{k}$$

$$\overline{BC} = \bar{i} + \bar{j} - 2\bar{k}$$

So,

$$\begin{aligned}\overline{AB} \times \overline{AC} &= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ -2 & 1 & 1 \\ -1 & 2 & -1 \end{vmatrix} \\ &= \bar{i}(-1-2) - \bar{j}(2+1) + \bar{k}(-4+1) \\ &= -3\bar{i} - 3\bar{j} - 3\bar{k}\end{aligned}$$

This implies,

$$|\overline{AB} \times \overline{AC}| = 3\sqrt{3}$$

And,

$$|\overline{BC}| = \sqrt{6}$$

Then,

$$h = \frac{3\sqrt{3}}{\sqrt{2}\sqrt{3}}$$

$$= \frac{3}{\sqrt{2}}$$

33. The volume of tetrahedron OABC is given by,

$$OABC = \frac{1}{6} [\bar{a} \bar{b} \bar{c}]$$

Now the position vectors are,

$$\frac{\bar{a} + \bar{b}}{3}, \frac{\bar{b} + \bar{c}}{3}, \frac{\bar{c} + \bar{a}}{3}, \frac{\bar{a} + \bar{b} + \bar{c}}{3}$$

So, conterminous edge vectors of new tetrahedron are $\frac{\bar{a}}{3}, \frac{\bar{b}}{3}, \frac{\bar{c}}{3}$

So, the volume of new tetrahedron is,

$$\frac{1}{6} \left[\frac{\bar{a}}{3} \frac{\bar{b}}{3} \frac{\bar{c}}{3} \right] = \frac{1}{6 \times 27} [\bar{a} \bar{b} \bar{c}]$$

So, the required ratio is $\frac{1}{27}$.

34. From the given vectors,

$$\begin{aligned}
 \vec{A} \times \vec{B} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} \\
 &= \vec{i}(2) - \vec{j}(2) + \vec{k}(2-1) \\
 &= 2\vec{i} - 2\vec{j} + \vec{k}
 \end{aligned}$$

This implies,

$$\begin{aligned}
 |(\vec{A} \times \vec{B}) \times \vec{C}| &= |(\vec{A} \times \vec{B})| |\vec{C}| \sin 30^\circ \\
 &= \sqrt{4+4+1} |\vec{C}| \left(\frac{1}{2}\right) \\
 &= \frac{3}{2} |\vec{C}| \\
 |\vec{C} - \vec{A}| &= 2\sqrt{2}
 \end{aligned}$$

This implies,

$$\begin{aligned}
 |\vec{C}|^2 + |\vec{A}|^2 - 2\vec{C} \cdot \vec{A} &= 8 \\
 |\vec{C}|^2 + 9 - 2|\vec{C}| &= 8 \\
 (|\vec{C}| - 1)^2 &= 0 \\
 |\vec{C}| &= 1
 \end{aligned}$$

So,

$$|(\vec{A} \times \vec{B}) \times \vec{C}| = \frac{3}{2}$$

35. The mean of given data is,

$$\begin{aligned}\bar{x} &= \frac{a_0 + a_1 + a_2 + \dots + a_{11}}{12} \\ &= \frac{a_0 + a_{11}}{2} \\ &= \frac{a_1 + a_{10}}{2} \\ &= \dots\end{aligned}$$

Then the deviation from mean is given by,

$$\begin{aligned}|\bar{x} - a_0| &= \frac{|a_{11} - a_0|}{2} \\ &= \frac{11|d|}{2}\end{aligned}$$

And,

$$\begin{aligned}|\bar{x} - a_1| &= \frac{|a_{10} - a_1|}{2} \\ &= \frac{9|d|}{2}, \dots\end{aligned}$$

So, sum of deviations is,

$$2 \left[\frac{11|d|}{2} \cdot 6 + \frac{9|d|}{2} + \frac{7|d|}{2} + \frac{5|d|}{2} + \frac{3|d|}{2} + \frac{|d|}{2} \right] = 36|d|$$

The mean deviation from arithmetic mean is given by,

$$\frac{36|d|}{12} = 3|d|$$

36. Consider given data,

Class Interval	Frequency (f_i)	x_i	$x_i f_i$	$(\bar{x} - x_i)^2$	$f_i (\bar{x} - x_i)^2$
0-10	2	5	10	196	392
10-20	3	5	45	16	48
20-30	4	25	100	36	144
30-40	1	35	35	256	256
$\sum f_i = 10$		$\sum x_i f_i = 190$ $\sum f_i (\bar{x} - x_i)^2 = 840$			

In the above table,

$$\begin{aligned}\bar{x} &= \frac{\sum x_i f_i}{N} \\ &= \frac{190}{10} \\ &= 19\end{aligned}$$

Thus the variance is,

$$\begin{aligned}\sigma^2 &= \frac{1}{N} \sum f_i (\bar{x} - x_i)^2 = \frac{1}{10}(840) \\ &= 84\end{aligned}$$

37. The required probability is given by,

$$\begin{aligned}\frac{{}^{73}C_{28} + {}^{73}C_{43}}{{}^{75}C_{30}} &= \frac{\frac{73!}{28!45!} + \frac{73!}{43!30!}}{\frac{75!}{30!45!}} \\ &= \frac{\frac{1}{45(44)} + \frac{1}{30(29)}}{\frac{75(74)}{(30)(29)(45)(44)}} \\ &= \frac{2(29) + (44)(3)}{5(74)} \\ &= \frac{19}{37}\end{aligned}$$

38. From the given data,

$$\frac{(n-1)(1) + (n+1)\frac{1}{2}}{2n} = \frac{41}{56}$$

$$\frac{2n-2+n+1}{4n} = \frac{41}{56}$$

$$42n-14=41n$$

$$n=14$$

So, number of unfair coins in the bag is given by,

$$n-1=14-1$$

$$=13$$

39. The required probability is given by,

$$\frac{\frac{1}{4} \left(\frac{{}^6C_2 + {}^8C_2}{{}^{14}C_2} \right)}{\frac{1}{4} \left(\frac{{}^6C_2 + {}^8C_2}{{}^{14}C_2} \right) + \frac{3}{4} \left(\frac{{}^9C_2 + {}^5C_2}{{}^{14}C_2} \right)} = \frac{(6)(5) + (8)(7)}{((6)(5) + (8)(7)) + 3[(9)(8) + (5)(4)]}$$

$$= \frac{86}{86 + 276}$$

$$= \frac{43}{181}$$

40. Consider given data,

$X = x_i$	1	2	3	4	5	6
$P(X = x_i)$	0.2	0.3	0.12	0.1	0.2	0.08

Then,

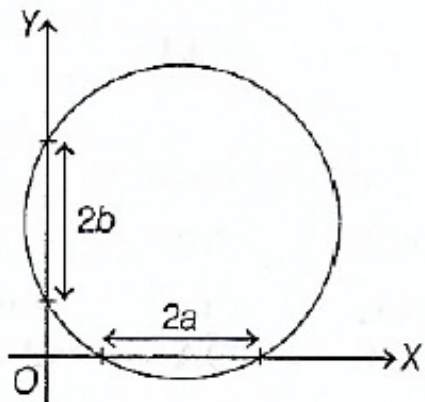
Now, $A = \{2, 3, 5\}$ and $B = \{1, 2, 3\}$.

$$\begin{aligned}
 P(A \cup B) &= P(X=1) + P(X=2) + P(X=3) + P(X=5) \\
 &= 0.2 + 0.3 + 0.12 + 0.2 \\
 &= 0.82
 \end{aligned}$$

41. For the unit mean,

$$\begin{aligned}
 \sum_{x=0}^{\infty} |x - \bar{x}| P(X = x) &= \sum_{x=0}^{\infty} |x - \bar{x}| \frac{e^{-1}}{x!} \\
 &= \frac{1}{e} \left[1 + \sum_{x=1}^{\infty} \frac{1}{(x-1)!} - \sum_{x=1}^{\infty} \frac{1}{x!} \right] \\
 &= \frac{1}{e} [1 + e - (e - 1)] \\
 &= \frac{2}{e}
 \end{aligned}$$

42. Consider the diagram,



Let the equation of circle be,

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

This implies,

$$2\sqrt{g^2 - c} = 2a$$

And,

$$2\sqrt{f^2 - c} = 2b$$

Then,

$$g^2 - a^2 = 0 \text{ and } f^2 - b^2 = 0$$

So,

$$g^2 - a^2 = f^2 - b^2$$

$$g^2 - f^2 = a^2 - b^2$$

When center is locus $(-g, -f)$ then,,

$$x^2 - y^2 = a^2 - b^2$$

43. After the rotation of the coordinate axes with given angle the relation between new (X, Y) and old coordinates (x, y) is given by,

$$(x, y) = [X \cos \theta - Y \sin \theta, (Y \cos \theta + X \sin \theta),$$

The angle is $\theta = \frac{\pi}{4}$.

Then,

$$(x, y) = \left(\frac{X}{\sqrt{2}} - \frac{Y}{\sqrt{2}}, \left(\frac{Y}{\sqrt{2}} + \frac{X}{\sqrt{2}} \right) \right)$$

So,

$$25 \left(\frac{X - Y}{\sqrt{2}} \right)^2 + 9 \left(\frac{X + Y}{\sqrt{2}} \right)^2 = 225$$

$$34X^2 + 34Y^2 - 32XY = 450$$

$$17X^2 + 17Y^2 - 16XY = 225$$

Comparison of the coefficients gives,

$$\alpha = \gamma = 17, \beta = -16, \delta = 225$$

This implies,

$$\begin{aligned}(\alpha + \beta + \gamma - \sqrt{\delta})^2 &= (34 - 16 - 15)^2 \\&= 3^2 \\&= 9\end{aligned}$$

44. The equation of line through the intersection of given lines is

$$\begin{aligned}(3x - 4y + 1) + \lambda(5x + y - 1) &= 0 \\(3 + 5\lambda)x + (\lambda - 4)y &= (\lambda - 1) \\\frac{x}{\frac{\lambda - 1}{3 + 5\lambda}} + \frac{y}{\frac{\lambda - 1}{\lambda - 4}} &= 1\end{aligned}$$

From the given information,

$$\frac{\lambda - 1}{3 + 5\lambda} = \frac{\lambda - 1}{\lambda - 4} \text{ and } \lambda \neq 1.$$

Then,

$$\lambda - 4 = 5\lambda + 3$$

$$4\lambda = -7$$

$$\lambda = -\frac{7}{4}$$

So, equation of the required line is,

$$\begin{aligned}\left(3 - \frac{35}{4}\right)x + \left(-\frac{7}{4} - 4\right)y &= \left(-\frac{7}{4} - 1\right) \\ -\frac{23}{4}x - \frac{23}{4}y &= -\frac{11}{4} \\ 23x + 23y &= 11\end{aligned}$$

45. The equation of the line that passes through $P(a, 2)$ and $a \neq 0$ making an angle 45° with positive X-axis is,

$$\frac{x-a}{\frac{1}{\sqrt{2}}} = \frac{y-2}{\frac{1}{\sqrt{2}}} = r$$

$$x = a + \frac{r}{\sqrt{2}} \text{ and } y = 2 + \frac{r}{\sqrt{2}}$$

So,

$$r = -a\sqrt{2}$$

For point C,

$$\begin{aligned}PB &= \sqrt{4+4} \\ &= 2\sqrt{2}\end{aligned}$$

And,

$$\begin{aligned} PC &= \sqrt{a^2 + a^2} \\ &= a\sqrt{2} \end{aligned}$$

From the points A and D,

$$\begin{aligned} \frac{\left(a + \frac{r}{\sqrt{2}}\right)^2}{9} + \frac{\left(2 + \frac{r}{\sqrt{2}}\right)^2}{4} &= 1 \\ 4\left(a + \frac{r}{\sqrt{2}}\right)^2 + 9\left(2 + \frac{r}{\sqrt{2}}\right)^2 &= 36 \end{aligned}$$

This implies,

$$13\left(\frac{r^2}{2}\right) + \left(\frac{8a}{\sqrt{2}} + \frac{36}{\sqrt{2}}\right)r + 4a^2 = 0$$

The roots are given by,

$$\begin{aligned} r_1 r_2 &= \frac{4a^2}{13} \\ &= \frac{8a^2}{13} \\ &= (PA)(PD) \end{aligned}$$

This implies,

PA, PB, PC, PD are in G.P so,

$$(PA)(PD) = (PB)(PC)$$

$$\frac{8a^2}{13} = 4a$$

$$2a = 13$$

46. The point $B(x_1, y_1)$ is the reflection of point A with respect to the line $x - y + 5 = 0$ so,

$$\begin{aligned}\frac{x_1 - 1}{1} &= \frac{y_1 + 2}{-1} \\ &= -2 \left(\frac{1 + 2 + 5}{2} \right) \\ &= -8 \\ x_1 &= -7\end{aligned}$$

And,

$$y_1 = 6$$

So, point is $B(-7, 6)$.

Similarly, for point $C(x_2, y_2)$ is reflection of A w.r.t line

$$x + 2y = 0 \text{ so,}$$

$$\begin{aligned}\frac{x_2 - 1}{1} &= \frac{y_2 + 2}{2} \\ &= -2 \left(\frac{1 - 4}{5} \right) \\ &= \frac{6}{5} \\ x_2 &= \frac{11}{5}\end{aligned}$$

And,

$$y_2 = \frac{2}{5}$$

So, point C is $\left(\frac{11}{5}, \frac{2}{5} \right)$

So equation of line BC is,

$$y - 6 = \frac{6 - \frac{2}{5}}{-7 - \frac{11}{5}}(x + 7)$$

$$\begin{aligned}y - 6 &= \frac{28}{-46}(x + 7) \\ &= \frac{-14}{23}(x + 7)\end{aligned}$$

Further simplify the above,

$$14x + 23y = 138 - 98$$

$$14x + 23y - 40 = 0$$

47. From the given data,

$$\frac{|4 - 3m|}{\sqrt{1 + m^2}} = 4$$

$$16 + 9m^2 - 24m = 16 + 16m^2$$

$$7m^2 + 24m = 0$$

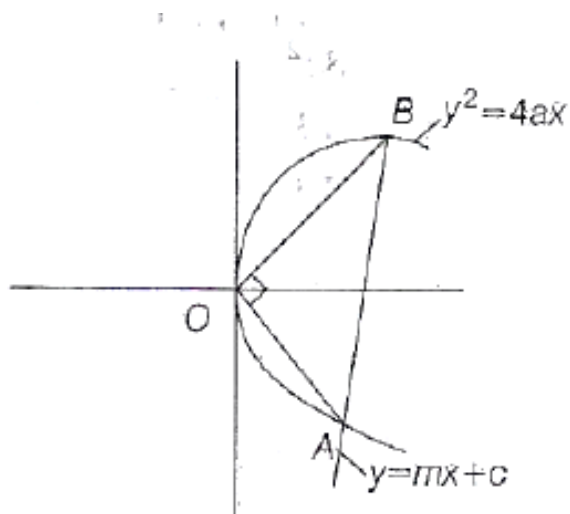
$$m = 0, -\frac{24}{7}$$

So, combined equation of the required line is,

$$y\left(y + \frac{24}{7}x\right) = 0$$

$$7y^2 + 24xy = 0$$

48. Consider the diagram,



Then,

$$y^2 - 4ax \left(\frac{y - mx}{c} \right) = 0$$

$$c + 4am = 0 \quad (I)$$

On substituting value of c it gives,

$$y = m(x - 4a)$$

It represents the family of lines passing through $(4a, 0)$.

49. Since it is given that abscissae of two points P and Q are roots of the equation $2x^2 + 4x - 7 = 0$ and their coordinates are the roots of the equation $3x^2 - 12x - 1 = 0$.

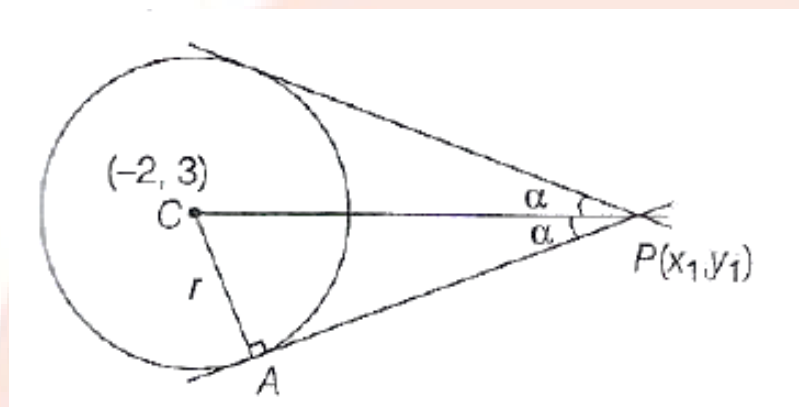
Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ then,

$$x_1 + x_2 = -2 \text{ and } y_1 + y_2 = 4.$$

So, the center of circle with PQ as diameter is,

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) = (-1, 2)$$

50. Consider the diagram,



Then,

$$\begin{aligned} \tan \alpha &= \frac{AC}{PA} \\ &= \frac{\sqrt{4 + 9 - 9\sin^2 \alpha + 13\cos^2 \alpha}}{\sqrt{x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9\sin^2 \alpha + 13\cos^2 \alpha}} \\ &= \frac{\sqrt{13\sin^2 \alpha - 9\sin^2 \alpha}}{\sqrt{x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9 + 4\cos^2 \alpha}} \\ &= \frac{\sqrt{4\sin^2 \alpha}}{\sqrt{x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9 + 4\cos^2 \alpha}} \end{aligned}$$

This implies,

$$\frac{\sin^2 \alpha}{\cos^2 \alpha} = \frac{4 \sin^2 \alpha}{x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9 + 4 \cos^2 \alpha}$$

$$x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9 + 4 \cos^2 \alpha = 4 \cos^2 \alpha$$

$$x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9 = 0$$

When taking locus of point $P(x_1, y_1)$, we get

$$x^2 + y^2 + 4x - 6y + 9 = 0$$

51. Consider the equation,

$$x^2 + y^2 - 4x - 6y - 12 = 0$$

The center of above is $C_1(2, 3)$ and radius is,

$$\begin{aligned} r_1 &= \sqrt{4 + 9 + 12} \\ &= 5 \end{aligned}$$

Let the center of required circle be $C_2(h, k)$ and radius is given as 3 touches given circle as $A(-1, -1)$.

The point $A(-1, -1)$ divides the line joining the centers $C_1(2, 3)$ and $C_2(h, k)$ externally in 5:3 so,

$$\begin{aligned}(-1, -1) &= \left(\frac{5h - 3(2)}{5 - 3}, \frac{5k - 3(3)}{5 - 3} \right) \\ &= \left(\frac{5h - 6}{2}, \frac{5k - 9}{2} \right)\end{aligned}$$

From above,

$$5h - 6 = -2 \text{ and } 5k - 9 = -2.$$

This implies,

$$h = \frac{4}{5} \text{ and } k = \frac{7}{5}$$

So, equation of required circle is,

$$\left(x - \frac{4}{5} \right)^2 + \left(y - \frac{7}{5} \right)^2 = 3^2$$

$$x^2 + y^2 - \frac{8x}{5} - \frac{14y}{5} + \frac{65}{25} = 9$$

$$x^2 + y^2 - \frac{8x}{5} - \frac{14y}{5} + \frac{13}{5} = 9$$

$$5x^2 + 5y^2 - 8x - 14y - 32 = 0$$

52. Consider the equation of circle,

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (\text{I})$$

When center is $(-g, -f)$ then it must lie on the line,

$$2x + 3y - 7 = 0$$

$$2g + 3f + 7 = 0 \quad (II)$$

Since equation (I) cuts the given circle so,

$$x^2 + y^2 - 4x - 6y + 11 = 0$$

$$x^2 + y^2 - 10x - 4y + 21 = 0$$

Orthogonally,

$$2g(-2) + 2f(-3) = c + 11$$

$$4g + 6f + c + 11 = 0 \quad (III)$$

And,

$$2g(-5) + 2f(2) = c + 21$$

$$10g + 4f + c + 21 = 0 \quad (IV)$$

From equation (III) and (IV),

$$6g - 2f + 10 = 0 \quad (V)$$

From equation (II) and (V),

$$11f + 11 = 0$$

$$f = -1$$

So,

$$g = -2, c = 3$$

Then,

$$\begin{aligned}
 5g - 10f + 3c &= 5(-2) - 10(-1) + 33 \\
 &= -10 + 10 + 9 \\
 &= 9
 \end{aligned}$$

53. Consider the equation of circle,

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

And,

$$2x^2 + 2y^2 + 3x + 8y + 2c = 0$$

The radical axis of the above circles is,

$$(4g - 3)x + (4f - 8)y = 0$$

The radical axis touches the circle,

$$x^2 + y^2 + 2x + 2y + 1 = 0$$

So,

$$\frac{|-(4g - 3) - (4f - 8)|}{\sqrt{(4g - 3)^2 + (4f - 8)^2}} = \sqrt{1 + 1 - 1}$$

$$(4g - 3)^2 + (4f - 8)^2 + 2(4g - 3)(4f - 8) = (4g - 3)^2 + (4f - 8)^2$$

$$8(4g - 3)(f - 2) = 0$$

$$(4g - 3)(f - 2) = 0$$

54. The length of the chord intercepted from line $y = mx + c$ by parabola $x^2 = 4ay$ is,

$$4\sqrt{a(1+m^2)(c+am^2)}$$

So,

$$\sqrt{40} = 4\sqrt{a(1+4)(1+a(4))}$$

$$40 = 16(5a(1+4a))$$

$$1 = 2a(1+4a)$$

$$8a^2 + 2a - 1 = 0$$

Further simplify the above,

$$8a^2 + 4a - 2a - 1 = 0$$

$$4a(2a+1) - 1(2a+1) = 0$$

$$a = \frac{1}{4} \text{ or } -\frac{1}{2}$$

$$4a = 1 \text{ or } -2$$

55. The equation of the normal that have slope m to the parabola $y^2 = 4ax$ is,

$$y = mx - 2am - am^3$$

It passes through (h, k) then,

$$k = mh - 2am - am^3$$

Let the roots be m_1, m_2, m_3 so,

$$m_1 m_2 m_3 = -\frac{k}{a}$$

And,

$$m_1 m_2 = -1$$

So,

$$m_3 = \frac{k}{a}$$

Then,

$$k = \frac{k}{a}h - 2a\frac{k}{a} - a\left(\frac{k}{a}\right)^3$$

$$1 = \frac{h}{a} - 2 - \frac{k^2}{a^2}$$

$$\frac{k^2}{a^2} = \frac{h}{a} - 3$$

$$k^2 = a(h - 3a)$$

By taking locus of (h, k) ,

$$y^2 - ax + 3a^2 = 0$$

56. Consider the ellipse,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Let $P(a \cos \theta, b \sin \theta)$ be the point of above ellipse.

The area of ΔPF_1F_2 is given by,

$$\begin{aligned}\Delta PF_1F_2 &= \frac{1}{2}(2ae)b|\sin \theta| \\ &= aeb|\sin \theta| \\ &= A\end{aligned}$$

For the maximum value of A , $\theta = \frac{\pi}{2}$ or $\frac{3\pi}{2}$ so,

$$A_{\max} = aeb$$

57. The equation of chord that joins $A(\alpha)$ and $B(\beta)$ on the ellipse,

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

Is,

$$\frac{x}{5} \cos \frac{\alpha + \beta}{2} + \frac{y}{3} \sin \frac{\alpha + \beta}{2} = \cos \frac{\alpha - \beta}{2} \quad (I)$$

Above equation passes through $(4,0)$ so,

$$\frac{4}{5} \cos \frac{\alpha + \beta}{2} = \cos \frac{\alpha - \beta}{2}$$

$$4 \left(\cos \frac{\alpha}{2} \cos \frac{\beta}{2} - \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \right) = 5 \left(\cos \frac{\alpha}{2} \cos \frac{\beta}{2} + \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \right)$$

$$4 \left(\cot \frac{\alpha}{2} \cot \frac{\beta}{2} - 1 \right) = 5 \left(\cot \frac{\alpha}{2} \cot \frac{\beta}{2} + 1 \right)$$

$$\cot \frac{\alpha}{2} \cot \frac{\beta}{2} = -9$$

58. Consider the equation.

$$16x^2 - 25y^2 - 96x + 100y - 356 = 0$$

$$\frac{(x-3)^2}{25} - \frac{(y-2)^2}{16} = 1$$

The equation of the tangent to above with slope 1 is,

$$y - 2 = 1(x - 3) + \sqrt{25(1) - 16}$$

$$y - 2 = x - 3 + 3$$

$$x - y + 2 = 0$$

59. Consider the given points,

$$PQ = \sqrt{1+1+16}$$

$$= 3\sqrt{2}$$

And,

$$\begin{aligned}QR &= \sqrt{9+9+0} \\ &= 3\sqrt{2}\end{aligned}$$

And

$$\begin{aligned}PR &= \sqrt{16+4+16} \\ &= \sqrt{36} \\ &= 6\end{aligned}$$

This implies,

$$\begin{aligned}PQ^2 + QR^2 &= 18 + 18 \\ &= 36 \\ &= PR^2\end{aligned}$$

Also, $PQ = QR$

This implies, PQR is a right angled isosceles triangle.

60. Consider given points,

$$A(2,3,5), B(\alpha,3,3), C(7,5,\beta)$$

The mid-point of BC is,

$$D\left(\frac{\alpha+7}{2}, 4, \frac{3+\beta}{2}\right)$$

The direction ratios of the lines joining the points A and D is,

$$\left(\frac{\alpha+3}{2}, 1, \frac{\beta-7}{2}\right)$$

The line axed AD is equally inclined with coordinate axis is,

$$\frac{\alpha + 3}{2} = 1 = \frac{\beta - 7}{2}$$

This implies,

$$\alpha = -1 \text{ and } \beta = 9.$$

So,

$$\cos^{-1}\left(\frac{\alpha}{\beta}\right) = \cos^{-1}\left(-\frac{1}{9}\right)$$

61. The equation of the plane that passes through origin and given line is,

$$(\bar{r} - 0) \cdot \left[(\bar{i} + 2\bar{j} - 3\bar{k}) \times (2\bar{i} - 3\bar{j} + \bar{k}) \right] = 0$$

$$\bar{r} \cdot [\bar{i}(2 - 9) - \bar{j}(1 + 6) + \bar{k}(-3 - 4)] = 0$$

$$\bar{r} \cdot [-7\bar{i} - 7\bar{j} - 7\bar{k}] = 0$$

$$\bar{r} \cdot [\bar{i} + \bar{j} + \bar{k}] = 0 \text{ or } x + y + z = 0$$

62. Consider the given expression,

$$\lim_{x \rightarrow \infty} \left\{ \frac{x^3 + 1}{x^2 + 1} - (\alpha x + \beta) \right\} = 2$$

Then,

$$\lim_{x \rightarrow \infty} \left\{ \frac{x^3 + 1 - \alpha x^3 + \beta x^2 - \alpha x - \beta}{x^2 + 1} \right\} = 2$$

For the above limit to exist, $x^3 = 0$ so,

$$\alpha = 1$$

Then,

$$\lim_{x \rightarrow \infty} \frac{-\beta x^2 - x - \beta + 1}{x^2 + 1} = 2$$

$$\lim_{x \rightarrow \infty} \frac{-\beta - \frac{1}{x} - \frac{\beta}{x^2} + \frac{1}{x^2}}{1 + \frac{1}{x^2}} = 2$$

$$-\beta = 2$$

$$\beta = -2$$

This implies,

$$(\alpha, \beta) = (1, -2)$$

63. Consider the expression for $k > 0$,

$$\sum_{x=0}^{\infty} \frac{k^x}{x!} \lim_{x \rightarrow \infty} \frac{n!}{(n-x)!} \left(1 - \frac{k}{n}\right)^{n-x} \left(\frac{1}{n}\right)^x$$

Simplify the above,

$$\begin{aligned}
\lim_{n \rightarrow \infty} \sum_{x=0}^{\infty} \frac{n!}{(n-x)!} \left(1 - \frac{k}{n}\right)^{n-x} \left(\frac{k}{n}\right)^x &= \lim_{n \rightarrow \infty} \sum_{x=0}^{\infty} {}^n C_x \left(1 - \frac{k}{n}\right)^{n-x} \left(\frac{k}{n}\right)^x \\
&= \lim_{n \rightarrow \infty} \left(1 - \frac{k}{n} + \frac{k}{n}\right)^n \\
&= \lim_{n \rightarrow \infty} 1^n \\
&= 1
\end{aligned}$$

64. Consider the function,

$$f(x) = \begin{cases} 5, & \text{if } x \leq 1 \\ a + bx, & \text{if } 1 < x < 3 \\ b + 5x, & \text{if } 3 \leq x < 5 \\ 30, & \text{if } x \geq 5 \end{cases}$$

If $f(x)$ is continuous at $x = 1$ then,

$$a + b = 5 \quad \text{(I)}$$

If continuous at $x = 3$ then,

$$\begin{aligned}
a + 3b &= b + 15 \\
a + 2b &= 15 \quad \text{(II)}
\end{aligned}$$

If continuous at $x = 5$ then,

$$\begin{aligned}
b + 25 &= 30 \\
b &= 5
\end{aligned}$$

Substitute 5 for b in equation (II).

$$a + 2(5) = 15$$

$$a = 5$$

But equation (I) is not satisfied by $a = 5$ and $b = 5$ then,

$f : R \rightarrow R$ is not continuous for any value of a and b .

65. Consider the function,

$$y = [x] + |1 - x|$$

The points of discontinuity are $x = 0, 1, 2, 3$ in $-1 \leq x \leq 3$.

So, function is not differentiable at 4 points.

66. Consider the equation,

$$\sqrt{1 - x^6} + \sqrt{1 - y^6} = a(x^3 - y^3)$$

$$\frac{\sqrt{1 - x^6} + \sqrt{1 - y^6}}{x^3 - y^3} = a$$

On differentiation both sides w.r.t x ,

$$\frac{\left[\left(x^3 - y^3 \right) \left(\frac{-6x^5}{2\sqrt{1-x^6}} - \frac{-6y^5}{2\sqrt{1-y^6}} \frac{dy}{dx} \right) \right] - \left(\sqrt{1-x^6} \right) + \left(\sqrt{1-y^6} \right) \left(3x^2 - 3y^2 \frac{dy}{dx} \right)}{(x^3 - y^3)^2} = 0$$

$$y^2 \frac{dy}{dx} \left[\frac{\left(\sqrt{1-x^6} + \sqrt{1-y^6} \right) + (1-y^6) - y^3 x^3 + y^6}{\sqrt{1-y^6}} \right] = x^2 \left[\frac{\left(\sqrt{1-x^6} + \sqrt{1-y^6} \sqrt{1-x^6} - y^3 x^3 + x^6 \right)}{\sqrt{1-x^6}} \right]$$

$$y^2 \frac{dy}{dx} = x^2 \sqrt{\frac{1-y^6}{1-x^6}}$$

67. Consider the expression,

$$\begin{aligned}\frac{d^2x}{dy^2} &= \frac{d}{dy} \left(\frac{dx}{dy} \right) \\ &= \frac{dx}{dy} \frac{d}{dx} \left(\frac{1}{\frac{dy}{dx}} \right) \\ &= \left(\frac{dx}{dy} \right) \left(\frac{-\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx} \right)^2} \right) \\ &= -\frac{\left(\frac{d^2y}{dx^2} \right)}{\left(\frac{dy}{dx} \right)^3}\end{aligned}$$

This implies,

$$\begin{aligned}\left(\frac{d^2x}{dy^2} \right)_p &= -\frac{(-3)}{4^3} \\ &= \frac{3}{64}\end{aligned}$$

68. Consider the expression for time period of simple pendulum,

$$T = 2\pi \sqrt{\frac{L}{g}}$$

This implies,

$$\frac{\Delta T}{T} \times 100 = \frac{1}{2} \frac{\Delta L}{L}$$

$$\begin{aligned}\frac{\Delta L}{L} \% &= -\frac{2 \times 2}{24 \times 60} \times 100 \\ &= -\frac{10}{36} \\ &= -\frac{5}{18}\end{aligned}$$

69. Consider the function,

$$f(x) = \sum_{k=1}^n (x - a_k)^2$$

Then,

$$\begin{aligned}f(x) &= \sum_{k=1}^n (x^2 - 2xa_k + a_k^2) \\ &= nx^2 - 2x(a_1 + a_2 + a_3 + \dots + a_n) + (a_1^2 + a_2^2 + \dots + a_n^2)\end{aligned}$$

The minimum value at $x = -\frac{b}{2a}$ so,

$$\begin{aligned}
 x &= -\frac{-2(a_1 + a_2 + a_3 + \dots + a_n)}{2n} \\
 &= \frac{a_1 + a_2 + a_3 + \dots + a_n}{n} \\
 &= A
 \end{aligned}$$

70. Consider the function,

$$f(x) = x^{2m-1}(a-x)^{2n}$$

The Rolle's theorem is applicable on the above function in the interval $[0, a]$.

$$f'(x) = (2m-1)x^{2m-2}(a-x) - 2n(a-x)^{2n-1}x^{2m-1}$$

When $x = c$ then $m > 1, n > 1$ and $f'(c) = 0$ so,

$$(2m-1)c^{2m-2} = 2nc^{2m-1}(a-c)^{2n-1}$$

$$\frac{(2m-1)}{c} = \frac{2n}{a-c}$$

$$c = \frac{a(2m-1)}{2n+2m-1}$$

71. Consider the function,

$$f(x) = \begin{cases} 2^x + 1, & \text{for } x \in [-1, 0) \\ 1, & \text{for } x = 0 \\ 2^x - 1, & \text{for } x \in (0, 1] \end{cases}$$

The function is strictly increasing in $x \in (-1, 0)$ and $f(0^-) \rightarrow 2$ but $f(0) = 1$.

And in interval $x \in (0, 1)$ again it is strictly increasing but $f(0^+) = 0$.

So, the function is neither maximum nor minimum.

72. Consider the function,

$$\int \frac{x-1}{(x+1)\sqrt{x^3+x^2+x}} dx$$

Simplify the expression

$$\begin{aligned} \int \frac{x^2-1}{(x+1)^2 x \sqrt{1+\frac{1}{x}+x}} dx &= \int \frac{x^2-1}{\left(1+1+\frac{1}{x}+x\right) x^2 \sqrt{1+\frac{1}{x}+x}} dx \\ &= \int \frac{1-\frac{1}{x^2}}{1+\left(1+\frac{1}{x}+x\right) \sqrt{1+\frac{1}{x}+x}} dx \end{aligned}$$

$$\text{Let } 1 + \frac{1}{x} + x = t^2$$

Then,

$$\left(1 - \frac{1}{x^2}\right) dx = 2t dt$$

This implies,

$$\begin{aligned} \int -\frac{2t dt}{(1+t^2)t} &= 2 \int \frac{dt}{1+t^2} \\ &= 2 \tan^{-1}(t) + c \\ &= 2 \tan^{-1}\left(\sqrt{\frac{1+x+x^2}{x}}\right) + c \end{aligned}$$

73. Consider the integral,

$$I(x) = \int x^2 (\log x)^2 dx$$

Simplify the above using “by parts”.

$$\begin{aligned} I(x) &= \frac{x^3}{3} (\log x)^2 - \int \frac{x^3}{3} \frac{2 \log x}{x} dx \\ &= \frac{x^3}{3} (\log x)^2 - \frac{2}{3} \left[\frac{x^3}{3} \right] (\log x) - \int \frac{x^3}{3} \frac{1}{x} dx \\ &= \frac{x^3}{3} (\log x)^2 - \frac{2}{3} \left[\frac{x^3}{3} (\log x) - \frac{1}{3} \frac{x^3}{3} \right] + C \\ &= \frac{x^3}{27} \left[9(\log x)^2 - 6(\log x) + 2 \right] + C \end{aligned}$$

For $I(1) = 0$,

$$\frac{2}{27} + C = 0$$

$$C = -\frac{2}{27}$$

This implies,

$$I(x) = \frac{x^3}{27} \left[9(\log x)^2 - 6(\log x) + 2 \right] - \frac{2}{27}$$

74. Consider the given integral,

$$\begin{aligned} & \int \frac{x^5 dx}{(x^2 + x + 1)(x^6 + 1)(x^4 - x^3 + x - 1)} \\ &= \int \frac{x^5 dx}{(x^2 + x + 1)(x^6 + 1)(x - 1)(x^3 + 1)} \\ &= \int \frac{x^5 dx}{(x^6 + 1)(x^3 - 1)(x^3 + 1)} \\ &= \int \frac{x^5 dx}{(x^6 + 1)(x^6 - 1)} \end{aligned}$$

Let $x^6 = t$ then, $6x^5 dx = dt$

This implies,

$$\begin{aligned} \int \frac{x^5 dx}{(x^6 + 1)(x^6 - 1)} &= \frac{1}{6} \int \frac{dt}{(t + 1)(t - 1)} \\ &= \frac{1}{12} \log_e \left| \frac{t - 1}{t + 1} \right| + c \\ &= \frac{1}{12} \log_e \left| \frac{x^6 - 1}{x^6 + 1} \right| + c \end{aligned}$$

75. Consider the integral,

$$I = \int \frac{dx}{x + \sqrt{x - 1}}$$

Take $x - 1 = t^2$ then, $dx = 2t dt$

Substitute these values in the integral and solve.

$$\begin{aligned} I &= \int \frac{2t}{(t^2 + 1) + t} dt \\ &= \int \frac{(2t + 1) - 1}{t^2 + t + 1} dt \\ &= \int \frac{2t + 1}{t^2 + t + 1} dt - \int \left(\frac{dt}{t^2 + t + 1} \right) \\ &= \log_e |t^2 + t + 1| - \int \frac{dt}{\left(t + \frac{1}{2}\right)^2 + \frac{3}{4}} \end{aligned}$$

Further simplify the above,

$$\begin{aligned} I &= \log_e |t^2 + t + 1| - \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + c \\ &= \log_e |t^2 + t + 1| - \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2t + 1}{\sqrt{3}} \right) + c \end{aligned}$$

Substitute the value of $t = \sqrt{x - 1}$.

$$I = \log_e |x + \sqrt{x - 1}| - \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2\sqrt{x - 1} + 1}{\sqrt{3}} \right) + c$$

76. Consider the given integral,

$$\int_{\log_e 2}^x \frac{dt}{\sqrt{e^t - 1}} = \frac{\pi}{6}$$

$$\int_{\log_e 2}^x \frac{e^{-\frac{t}{2}}}{\sqrt{1 - \left(e^{-\frac{t}{2}}\right)^2}} dt = \frac{\pi}{6}$$

Take $e^{-\frac{t}{2}} = u$ then, $e^{-\frac{t}{2}} dt = -2du$

$$\text{At } t = \log_e 2, u = \frac{1}{\sqrt{2}}$$

$$\text{At } t = x, u = e^{-\frac{x}{2}}$$

Substitute these values in the above expression.

$$\int_{\frac{1}{\sqrt{2}}}^{e^{\frac{x}{2}}} \frac{-2du}{\sqrt{1-(u)^2}} = \frac{\pi}{6}$$

$$\sin^{-1}\left(e^{\frac{x}{2}}\right) - \frac{\pi}{4} = -\frac{\pi}{12}$$

$$\sin^{-1}\left(e^{\frac{x}{2}}\right) = \frac{\pi}{6}$$

$$e^{\frac{x}{2}} = \frac{1}{2}$$

$$-\frac{x}{2} = \log_e\left(\frac{1}{2}\right)$$

$$= -\log_e(2)$$

$$x = 2\log_e(2)$$

77. Consider the given integral.

$$I = \int_0^1 \frac{\log_e(1+x)}{1+x^2} dx$$

Take $x = \tan \theta$, then $dx = \sec^2 \theta d\theta$

Substitute in the above integral and change the limits accordingly.

$$\begin{aligned} I &= \int_0^{\frac{\pi}{4}} \frac{\log_e (1 + \tan \theta)}{\sec^2 \theta} \sec^2 \theta d\theta \\ &= \int_0^{\frac{\pi}{4}} \log_e (1 + \tan \theta) d\theta \end{aligned}$$

Now, using the property $\int_0^a f(x) dx = \int_0^a f(a-x) dx$,

$$\begin{aligned} I &= \int_0^{\frac{\pi}{4}} \log_e \left(1 + \tan \left(\frac{\pi}{4} - \theta \right) \right) d\theta \\ &= \int_0^{\frac{\pi}{4}} \log_e \left(1 + \frac{1 - \tan \theta}{1 + \tan \theta} \right) d\theta \\ &= \int_0^{\frac{\pi}{4}} \log_e \left(\frac{2}{1 + \tan \theta} \right) d\theta \\ &= \int_0^{\frac{\pi}{4}} \log_e (2) dx - \int_0^{\frac{\pi}{4}} \log_e (1 + \tan \theta) d\theta \end{aligned}$$

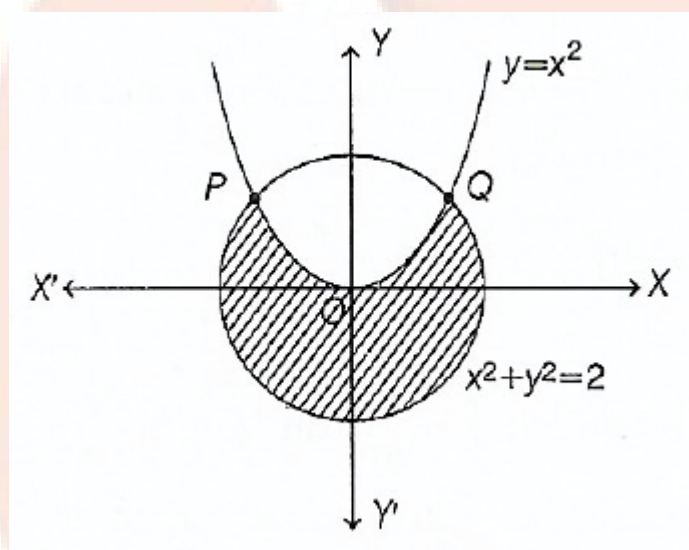
Further simplifying, we get

$$I = \frac{\pi}{4} \log_e(2) - I$$

$$2I = \frac{\pi}{4} \log_e(2)$$

$$I = \frac{\pi}{8} \log_e(2)$$

78. Consider the diagram,



Solve the given equations for the point of intersections.

$$y^2 + y - 2 = 0$$

$$y^2 + 2y - y - 2 = 0$$

$$y(y + 2) - 1(y + 2) = 0$$

$$y = 1 \quad [\because y > 0]$$

Thus, the area of the larger part will be,

$$\begin{aligned}\text{Area} &= \pi + 2 \int_0^1 \left(\sqrt{2-y^2} - \sqrt{y} \right) dy \\ &= \pi + 2 \left[\frac{y}{2} \sqrt{2-y^2} + \frac{2}{2} \sin^{-1} \frac{y}{\sqrt{2}} - \frac{2}{3} y^{\frac{3}{2}} \right]_0^1 \\ &= \pi + 2 \left[\frac{1}{2} + \frac{\pi}{4} - \frac{2}{3} \right] \\ &= \frac{3\pi}{2} - \frac{1}{3} \text{ sq. units}\end{aligned}$$

79. Consider the equation of family of curves.

$$\begin{aligned}x^2 &= c(y+c)^2 \\ x &= \sqrt{c}(y+c) \quad \dots\dots (1)\end{aligned}$$

After differentiating w.r.t to x , we get

$$\begin{aligned}1 &= \sqrt{c} \frac{dy}{dx} \\ \sqrt{c} &= \frac{dx}{dy}\end{aligned}$$

Now, from equation (1), we get

$$x = \left(\frac{dx}{dy} \right) \left[y + \left(\frac{dx}{dy} \right)^2 \right]$$

$$x \left(\frac{dy}{dx} \right)^3 = y \left(\frac{dy}{dx} \right)^2 + 1$$

$$x \left(\frac{dy}{dx} \right)^3 - y \left(\frac{dy}{dx} \right)^2 - 1 = 0$$

80. Simplify the given matrix.

$$\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix} = 0$$

$$f(x)f''(x) - (f'(x))^2 = 0$$

$$f(x)f''(x) = (f'(x))^2$$

$$\frac{f''(x)}{f'(x)} = \frac{f'(x)}{f(x)}$$

Integrate both the sides.

$$\int \frac{f''(x)}{f'(x)} dx = \int \frac{f'(x)}{f(x)} dx$$

$$\log_e (f'(x)) = \log_e (f(x)) + c \quad \dots\dots (1)$$

Since $f(0) = 1$ and $f'(0) = 2$ then,

$$c = \log_e 2$$

Substituting the value of c in equation (1), we get

$$\log_e (f'(x)) = \log_e (f(x)) + \log_e 2$$

$$f'(x) = 2f(x)$$

$$\frac{f'(x)}{f(x)} = 2$$

Integrating both sides w. r. t. x ,

$$\int \frac{f'(x)}{f(x)} dx = \int 2 dx$$

$$\log_e f(x) = 2x + c$$

For $f(0) = 1$, $c = 0$.

This gives,

$$\log_e f(x) = 2x$$

$$f(x) = e^{2x}$$

